

Augur: an α -domain variable-projection DOAS retrieval, and what its diagnostics reveal that standard fit reports do not

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Abstract. A DOAS fit reports a residual, chi-square, uncertainty and convergence flag, read as a measure of retrieval quality. We present Augur, an open retrieval that poses the fit in the extinction domain as a separable nonlinear least-squares problem, solves it by variable projection and is configured by instrument and campaign profiles. Its calibration references enter as data, so they can be perturbed, and the objective mapped, without reprocessing a campaign. We use it on one field case, a thermal-
5 dissociation broadband cavity-enhanced absorption spectrometer (Min et al., 2016; Nam et al., 2022) with two heated and one ambient-temperature cavity, to measure two gaps between fit report and data.

The first is error the fit cannot see. For the alkyl nitrate product, leave-one-out perturbation of the zero-air, reflectivity and etalon references gives, in the best-calibrated week, a structural uncertainty 1.28 to 2.29 times the reported per-scan sigma, and a total 1.63 to 2.50 times, without enlarging the residual. The second is information the pipeline holds and discards. The
10 ambient-temperature channel’s wavelength shift is not identifiable from ambient spectra, yet its fits report convergence. A time-base error introduced by the pipeline, a reflectivity reference nine hours out of alignment, displaced the product by 0.100 ppb while changing the residual and the reported uncertainty by two parts in a thousand. A multiplicative scale error, evident only against a co-located reference analyser, is invisible to both. We propose that each record carry its termination state, distance to the nearest calibration and reference provenance, and release a synthetic benchmark for other implementations.

15 1 Introduction

A DOAS retrieval returns more than a concentration. It returns a residual, a chi-square, a parameter covariance and a convergence flag, and these travel with the concentration into averaging, filtering, trend fits and comparisons with models and other instruments. They are read as a statement of trust: a small residual means the model described the spectrum, a converged
20 flag means the optimiser found the answer, and the reported 1σ means the concentration is known that well. Each reading is conditional, and the conditions are not reported alongside the numbers.

This paper asks two questions about that gap. First, how large are the errors that a fit report cannot see, when they are measured by perturbing the calibration chain and re-running the retrieval? Second, what information does the pipeline already hold, and discard, that would flag a failed retrieval? A third category emerges from the answers: errors that neither the fit report nor any of the fields we propose can detect (Sect. 6.5).

25 The first question has a partial answer in the DOAS literature. The uncertainty a DOAS fit reports is the propagation of the
residual scatter through the linear system, and it is correct only if the residual is white and the forward model complete. Stutz
and Platt (1996) showed that structure in the residual inflates the true error beyond the reported one, and the standard treatment
(Platt and Stutz, 2008) sets out the conditions explicitly. Practitioners respond with rules of thumb – multiply the reported error
by two, or by the ratio of the residual to the photon noise – and with sensitivity studies on the window, the polynomial order
30 and the set of cross sections.

That literature examines the spectral fit. The quantities the fit treats as given – the reference spectrum, the mirror reflectivity,
the instrument line shape – are usually characterised separately, assigned a calibration uncertainty and combined with the fit
uncertainty at the end. For a cavity-enhanced instrument this is a strong assumption. The reference and the reflectivity are time
series, measured intermittently and interpolated to every measurement, and the error they contribute to a record depends on
35 where that record sits relative to the nearest calibration. A campaign-mean calibration uncertainty, of the kind reported for the
effective path length in earlier instruments of the design used here (Min et al., 2016; Nam et al., 2022), does not describe it.

Intercomparisons and synthetic studies do not close this gap either. Intercomparisons such as CINDI-2 (Kreher et al., 2020)
measure the spread between codes, which bounds implementation error and has standardised settings, but codes that agree may
share a mistake, and without a truth a disagreement cannot say which is right. Synthetic studies supply that truth, but they test
40 the fit for given references, which are correct by construction.

Treating the calibration chain as part of the retrieval divides the shortfalls of a fit report into two kinds with different
remedies. The first is what the fit cannot know: contributions that leave the residual unchanged, which have to be measured
by perturbing the retrieval and added to the budget. The second is what the pipeline already knows and does not carry to the
product: whether the optimiser stopped at a bound, whether the nearest reflectivity measurement was fifteen minutes or eleven
45 hours away, whether a quality gate rejected the neighbouring calibration blocks. This information exists when the concentration
is computed and is then discarded, although keeping it would turn a silent failure into a flagged one.

We present Augur, an open retrieval built so that both kinds of question can be asked routinely. It poses the retrieval in
the extinction domain as a separable nonlinear least-squares problem and solves it by variable projection. The layout of the
raw data, the calibration flags and the fit configuration are read from instrument and campaign profiles (Appendix A), so the
50 same code runs other configurations. The diagnostics are demonstrated on one field case: a thermal-dissociation broadband
cavity-enhanced absorption spectrometer with two heated and one ambient-temperature cavity, measuring NO₂, total peroxy
nitrates and total alkyl nitrates over 53 days in 2026. Other configurations are not evaluated here.

We verify the retrieval against synthetic truth and against an independent DOAS implementation. We then measure the
structural terms it cannot report, which together exceed the reported sigma by a factor of 1.6 to 2.5 in the best-calibrated week,
55 and map the identifiability of its nonlinear parameters. We also describe a time-base error introduced by the pipeline itself:
twelve days of operational product were built from a reflectivity reference that a later correction had made stale. It displaced
the alkyl nitrate product by 0.100 ppb and the 300 °C channel's concentration by two reported standard deviations, while the
residual and the reported uncertainty changed by two parts in a thousand. We propose a small reporting convention whose
distance fields would have flagged it at the first reprocessing and whose provenance fields, once implemented, would name its

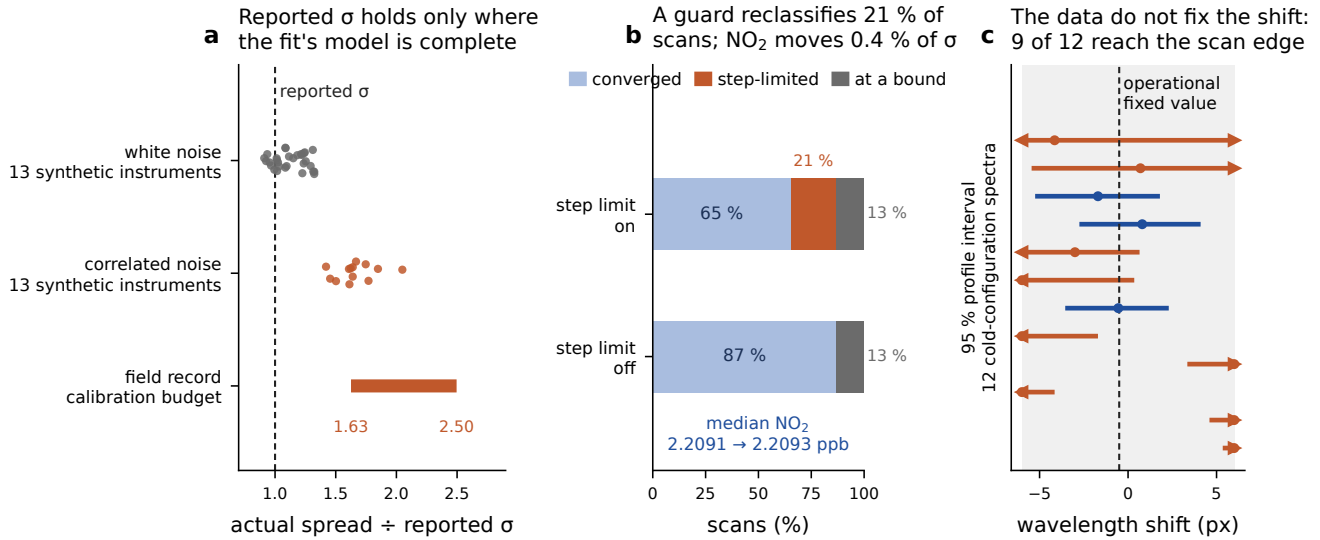


Figure 1. What a converged fit reports, and what the data support. **(a)** Spread of retrieved NO₂ over the reported uncertainty. Grey: white noise, 30 cases over 13 synthetic instruments (line-shape width 0.5–2, window 0.7–1.3 and pixel dispersion 0.7–1.4 times nominal; signal-to-noise over a factor of 32; polynomial orders 2–6). Orange: the same with AR(1) noise, $\rho = 0.5$. Bar: the field record, total uncertainty 1.63 (robust scale) to 2.50 (standard deviation) times the fit-propagated σ ($n = 8847$ records of 60 s). **(b)** Termination state of 8446 operational scans, step limit on and off. **(c)** 95 % profile-likelihood intervals of the cold-configuration shift for 12 spectra, scanned over ± 6 px, with $\text{RSS}/\text{RSS}_{\min} \leq 1 + F_{0.95}(1, n - P)/(n - P)$, $n - P = 766$; arrows reach the scan edge; dashed, the operational value, -0.5 px.

60 cause. Finally, we release a synthetic benchmark with which any DOAS implementation can measure the same quantities on the same cases.

The evidence is one field case, one instrument over one campaign, which bounds what can be concluded about instruments in general. Two results are intended to carry beyond it. The benchmark of Sect. 3.4 contains no retrieval code and no field data, so the measurements it supports do not depend on any implementation; its cross sections, windows and wavelength grid are those of one instrument. The reporting convention of Sect. 6 asks for quantities that every implementation already computes, apart from the build time and time convention of each reference, which must be recorded when the reference is built. Where a result depends on this instrument, such as the magnitude of the reflectivity term or the behaviour of the cold channel, we say so.

Figure 1 previews the argument. Section 2 sets out the retrieval and Sect. 3 verifies it. Sections 4 and 5 take the two kinds of shortfall in turn, Sect. 6 draws the reporting convention out of them and names what it cannot reach, Sect. 7 demonstrates it on the field case, and Sect. 8 concludes. Appendix A collects the instrument and retrieval configuration, Appendix B the projected Jacobian and joint covariance, Appendix C the measurements on calibration-block length, and Appendix D the choice of denominator for the budget ratio.

2 Retrieval formulation

75 2.1 Forward model in the extinction domain

The cavity output is converted to an extinction coefficient before any spectral fitting is attempted. For a sample spectrum $I(\lambda)$ and a reference spectrum $I_0(\lambda)$ recorded with the cavity filled by zero air,

$$\alpha(\lambda) = \left[\underbrace{\frac{R_L(1-R(\lambda,t))}{d}}_{\alpha_{\text{mirror}}} + R_L \alpha_{\text{Ray}}(\lambda; T_0, P_0) \right] \frac{I_0(\lambda) - I(\lambda)}{I(\lambda)} - [\alpha_{\text{Ray}}(\lambda; T, P) - \alpha_{\text{Ray}}(\lambda; T_0, P_0)], \quad (1)$$

where $R(\lambda, t)$ is the mirror reflectivity and d the mirror separation. R_L is the ratio of the full mirror separation to the cavity length actually filled by the sample, excluding the purge-gas volumes at the mirrors, and α_{Ray} is the Rayleigh extinction at the temperature and pressure of the sample (T, P) and of the reference (T_0, P_0) . The first bracket converts a relative intensity change into an absolute extinction; the second removes the Rayleigh difference between the two states. Light-level changes between reference and sample enter α directly, a sensitivity that the iterative cavity-enhanced DOAS method of Horbanski et al. (2019) removes. In α , unlike in optical density, the gas amounts enter linearly: the quantity to be fitted is a sum of reference cross sections with concentration coefficients, a smooth background and the periodic structure the optics impose.

2.2 Separable structure

Let p index the n detector pixels of the fitting window and let

$$x(p) = \frac{2(p - p_{\min})}{p_{\max} - p_{\min}} - 1 \in [-1, 1].$$

The model for the measured extinction is

$$90 \quad \alpha_{\text{model}}(p) = \sum_{i=1}^m c_i \tau_i(T) \sigma_i(p'_i) + \sum_{j=0}^q b_j T_j(x(p)) + a_s \sin(fp) + a_c \cos(fp), \quad (2)$$

$$p'_i = (p - p_c) s_i + p_c + \delta_i, \quad (3)$$

with T_j the Chebyshev polynomial of the first kind, p_c a fixed window centre, and $\tau_i(T) = 1 + \kappa_i(T - T_{\text{ref}})/100$ the temperature scaling of the i -th cross section. Each reference carries its own wavelength shift δ_i (pixels) and squeeze s_i ; in practice one gas is designated the anchor and the companions are linked to it or fixed, which constrains θ without changing the structure. The Chebyshev basis on $x \in [-1, 1]$ keeps the background block well conditioned, where a monomial basis would reach condition numbers that make the background coefficients meaningless and leak into the gas coefficients through the shared normal equations. The etalon enters as a sine–cosine pair at fixed frequency: since $A \sin(fp + \varphi) = A \cos \varphi \sin(fp) + A \sin \varphi \cos(fp)$, the pair (a_s, a_c) spans the same model space as (A, φ) but is linear, so only the frequency f is fixed a priori.

100 Collecting the $k = m + (q + 1) + 2$ basis vectors into a design matrix $\mathbf{A}(\theta)$ whose columns depend on the nonlinear vector $\theta = (\delta_i, s_i)$ only through Eq. (3), and writing $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$ for the pixel weights, the problem is

$$\min_{\theta, \mathbf{c}} \|\mathbf{W}(\mathbf{y} - \mathbf{A}(\theta) \mathbf{c})\|_2^2, \quad \mathbf{y} = \alpha_{\text{meas}}. \quad (4)$$

This is a separable nonlinear least-squares problem in the sense of Golub and Pereyra (1973): k linear parameters and, for the operational configuration, two to six nonlinear ones.

105 2.3 Variable projection with an analytic projected Jacobian

For any θ the inner problem is an ordinary linear least-squares problem, so $\mathbf{c}$ need never be searched. Substituting the minimiser $\hat{\mathbf{c}}(\theta) = \mathbf{A}^+ \mathbf{y}$ leaves the projected residual

$$\mathbf{r}(\theta) = (\mathbf{I} - \mathbf{A}(\theta)\mathbf{A}(\theta)^+) \mathbf{y} = \mathbf{P}_{\mathbf{A}(\theta)}^\perp \mathbf{y}, \quad (5)$$

a function of θ alone, which we minimise with a bounded Levenberg–Marquardt/trust-region solver. Its Jacobian is supplied
110 analytically in the full Golub–Pereyra form,

$$\frac{\partial \mathbf{r}}{\partial \theta_k} = - \left[\underbrace{\mathbf{P}^\perp \mathbf{D}_k \hat{\mathbf{c}}}_{\text{projection term}} + \underbrace{(\mathbf{A}^+)^T \mathbf{D}_k^T \mathbf{r}}_{\text{second term}} \right], \quad \mathbf{D}_k = \frac{\partial \mathbf{A}}{\partial \theta_k}. \quad (6)$$

Retaining the second term makes Eq. (6) the exact derivative rather than the Kaufman (1975) approximation, which can therefore be checked against finite differences to round-off (Sect. 3.2). The factorisation, the derivative of the gas columns, the invariance under column scaling and the relation to existing DOAS implementations, none of which forms this derivative, are
115 given in Appendix B5.

2.4 Time-resolved mirror reflectivity

R in Eq. (1) is determined at intervals from helium and zero-air injections, whose known Rayleigh extinction fixes the mirror loss (Washenfelter et al., 2008), and it drifts between them. We therefore carry $R(\lambda, t)$ as a monotone piecewise-cubic (PCHIP) interpolation (Fritsch and Carlson, 1980) through the injection knots, evaluated at the timestamp of each spectrum, rather than
120 as one value per campaign or per day. Monotone interpolation cannot overshoot between knots; the cost, quantified in Sect. 4.4, is that it also cannot represent a turning point that falls between two knots.

2.5 Uncertainty

With $\hat{\sigma}^2 = \mathbf{r}^T \mathbf{r} / (n - P)$, where P counts the linear columns and the active nonlinear parameters, the conditional covariance of the linear coefficients is the usual

$$125 \text{ Cov}(\mathbf{c} \mid \theta = \hat{\theta}) = \hat{\sigma}^2 (\mathbf{A}_{\mathbf{W}}^T \mathbf{A}_{\mathbf{W}})^{-1}. \quad (7)$$

Equation (7) is what DOAS retrievals conventionally report. It holds the fitted shift and squeeze at their optimum: variable projection separates θ from $\mathbf{c}$ in the solution but not in the uncertainty, and an uncertain shift passes its uncertainty to the concentrations. We therefore also form the joint covariance from the full Jacobian of the unprofiled residual,

$$\mathbf{J} = [\mathbf{A}_{\mathbf{W}} \mid \mathbf{V}], \quad \mathbf{V}_{:,k} = \sum_i \frac{\partial \mathbf{A}_{:,i}}{\partial \theta_k} \hat{c}_i, \quad \text{Cov}(\mathbf{c}, \theta) = \hat{\sigma}^2 (\mathbf{J}^T \mathbf{J})^{-1}, \quad (8)$$

130 and take the gas block (O’Leary and Rust, 2013). The joint standard errors are never smaller than the conditional ones (Appendix B4). Both quantities are reported, so that the size of the difference can be read directly. When $n - P \leq 0$ the variance estimate is reported as missing, and the scan is flagged as underdetermined and retained.

2.6 Termination state

A residual-and- χ^2 report cannot distinguish a fit that stopped at a parameter bound from one that stopped because the gradient
135 vanished. We therefore classify and export the termination state of the nonlinear stage separately from the solver’s own success flag: CONVERGED, AT_BOUND (a nonlinear parameter rests on its box constraint at exit), STEP_LIMITED (the trust-region step was capped), MAX_NFEV, and FAILED. The classifier is conservative by construction: a nonlinear parameter within a small tolerance of its box constraint (relative to the box width) is flagged even if the optimum lies there, and across the solves of one record the least favourable state is kept. It changes no retrieved value but changes what the record admits (Sects. 5.2
140 and 7.3). Two further implementation matters – the scaling of α before the solve and the non-negativity, Tikhonov and robust options, none of which is used operationally – are described in Appendix B6.

2.7 What the implementation exposes

Augur makes two questions cheap to ask of a finished retrieval. The first is what happens to the product when a calibration reference is changed. The structural budget of Sect. 4 removes one zero-air knot, one reflectivity knot or one etalon component
145 at a time and re-runs the retrieval, once per knot and then for the pairwise and triple combinations. It covers the 9034 records of the budget window that have a reflectivity knot within 1 h (8847 after bound-terminated fits are excluded). The second is what the objective looks like away from the converged point: the identifiability map of Sect. 5 evaluates the projected objective on a grid of 160 points in the wavelength shift for each of twelve spectra, and repeats a coarser version over 165. A fit report answers neither question; both require re-entering the retrieval with one input altered.

150 Both questions are loops over data already in memory. The time-resolved references enter the forward model as data, an array of knots with times interpolated at evaluation, so substituting one is an assignment, and the projected objective is a callable of the nonlinear parameters that can be evaluated at a chosen shift without converging a fit. In one process on the analysis workstation, the in-memory budget takes 0.74 s per record for both heated channels and all four perturbations (74 s for a 100-record subset), about 1.9 h for the 9034-record window. A file-based route that rebuilds the extinction record and
155 refits one channel-day (1399 records) takes 54 s and 90 s, about 4.8 min per perturbation for the two channels even if only the affected day is reprocessed. With 157 zero-air and 54 helium knots in the window, the single-knot removals alone would take about 17 h, before the pairwise and triple combinations. The formulation itself is conventional (Sect. 3.3); the contribution of the software is architectural, in that it brings the quantities of Sects. 4 and 5 to this cost. The nine-hour time-base error of Sect. 5.4, invisible in the residual, could be measured only by re-running the retrieval with the time base changed.

Table 1. Retrieval configurations used in Sects. 3–6, in short form; windows are rounded to the nanometre, and the full configuration, including the exact window bounds, is in Table A1.

Label	Cell	Window (nm)	Baseline	Shift range (px)	Light level
hot ANs	300 °C	430–462	4th order	−10 to 0.5	nominal
hot PNs	180 °C	444–471	3rd order	−5 to 5	nominal
cold	ambient	438–476	4th order	fixed at −0.5	reduced

160 **3 Verification**

The verification covers five properties. The implementation solves the equations of Sect. 2, the analytic Jacobian is the derivative it claims to be, and an independent implementation of the same problem returns the same answer. The diagnostics of Sects. 4 and 5 measure what they are said to measure on data whose truth is known, and the two hot channels share a relative scale. Absolute concentration accuracy against a reference standard is not established and is not claimed anywhere in this paper.

Results in Sects. 3–6 are reported for three retrieval configurations of the field case of Sect. 7, labelled by the cells they come from. Table 1 lists what distinguishes them for the retrieval: fitting window, baseline order, permitted shift range and light level; the full configuration is in Appendix A. None of the results depends on the chemistry the cells measure, and where a result depends on a property of a configuration, that property is named. Figure 2 shows one record (1 June 2026, 10:35 KST; cold channel 10:34:34) retrieved in all three configurations. Each window sits on the low-noise band set by its LED. The cold window extends 13 nm below its LED half-maximum, where the noise is several times higher. The step at 428 nm in Fig. 2a is the band-pass filter edge, and that at 470 nm in Fig. 2b the red edge of the 469 nm LED.

3.1 Recovery of a known truth

The retrieval reproduces its own forward model to machine precision. Spectra are generated from the forward model of Sect. 2.1 with prescribed gas amounts, a Chebyshev baseline, a wavelength shift and a squeeze, and retrieved with the same model. Over 27 combinations – nine shifts from −4 to +4 px (including sub-pixel values of ± 0.37 px) crossed with squeezes of 0.998, 1.000 and 1.002 – the largest error in the retrieved NO₂ amount is 9.5×10^{-13} relative and the largest error in the recovered shift 6.7×10^{-10} px. This is a statement about the solver, not about the instrument.

The baseline order is the one structural choice the retrieval makes on its own. Sweeping it from 0 to 8 on a noise-free spectrum built with a fourth-order baseline, the retrieved NO₂ is in error by 27 % at order 0 and 0.75 % at order 1, and reaches machine precision at order 2 and above. Across the sweep the normalised condition number rises from 14.9 to 33.8 and the degrees of freedom fall from 543 to 535. On noisy realisations of the same spectrum the scatter of the retrieved NO₂ is 0.88 to 0.91 times the reported 1σ at order 2 and above. For this configuration, with white photon noise and a correct model, the reported uncertainty is therefore accurate to about ten per cent, the baseline against which the structural terms of Sect. 4 should

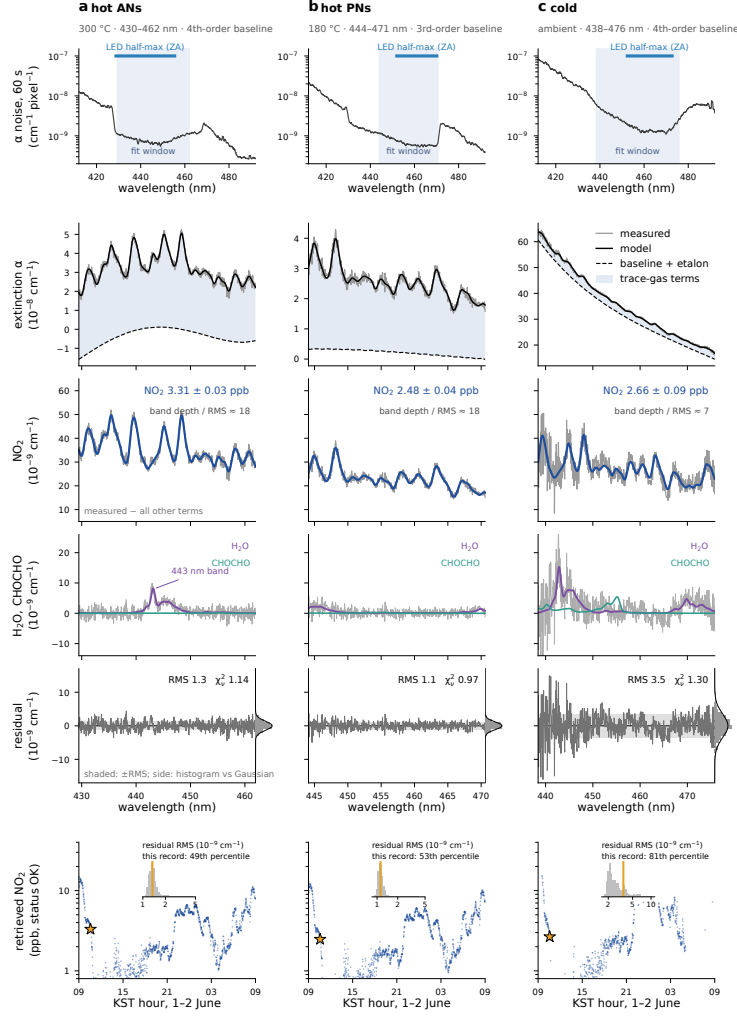


Figure 2. One record retrieved in the three configurations of Table 1: **(a)** hot ANs (300 °C), **(b)** hot PN (180 °C), **(c)** cold. Top: per-pixel noise of the 60 s extinction spectrum in the surrounding hour (fit window shaded; bar, LED half-maximum). Middle: extinction with fitted model and baseline-plus-etalon term; NO_2 and H_2O contributions (measurement minus other fitted terms) with their fits; residual with $\pm$ RMS and histogram. The NO_2 row gives the operational value, its 1σ , and peak-to-peak NO_2 over residual RMS. Bottom: NO_2 over the surrounding 24 h (reduced $\chi^2 \leq 1.5$; $n = 1397, 1399, 707$; star, this record); inset, residual RMS distribution, on which this record sits at the 49th, 53rd and 81st percentile.

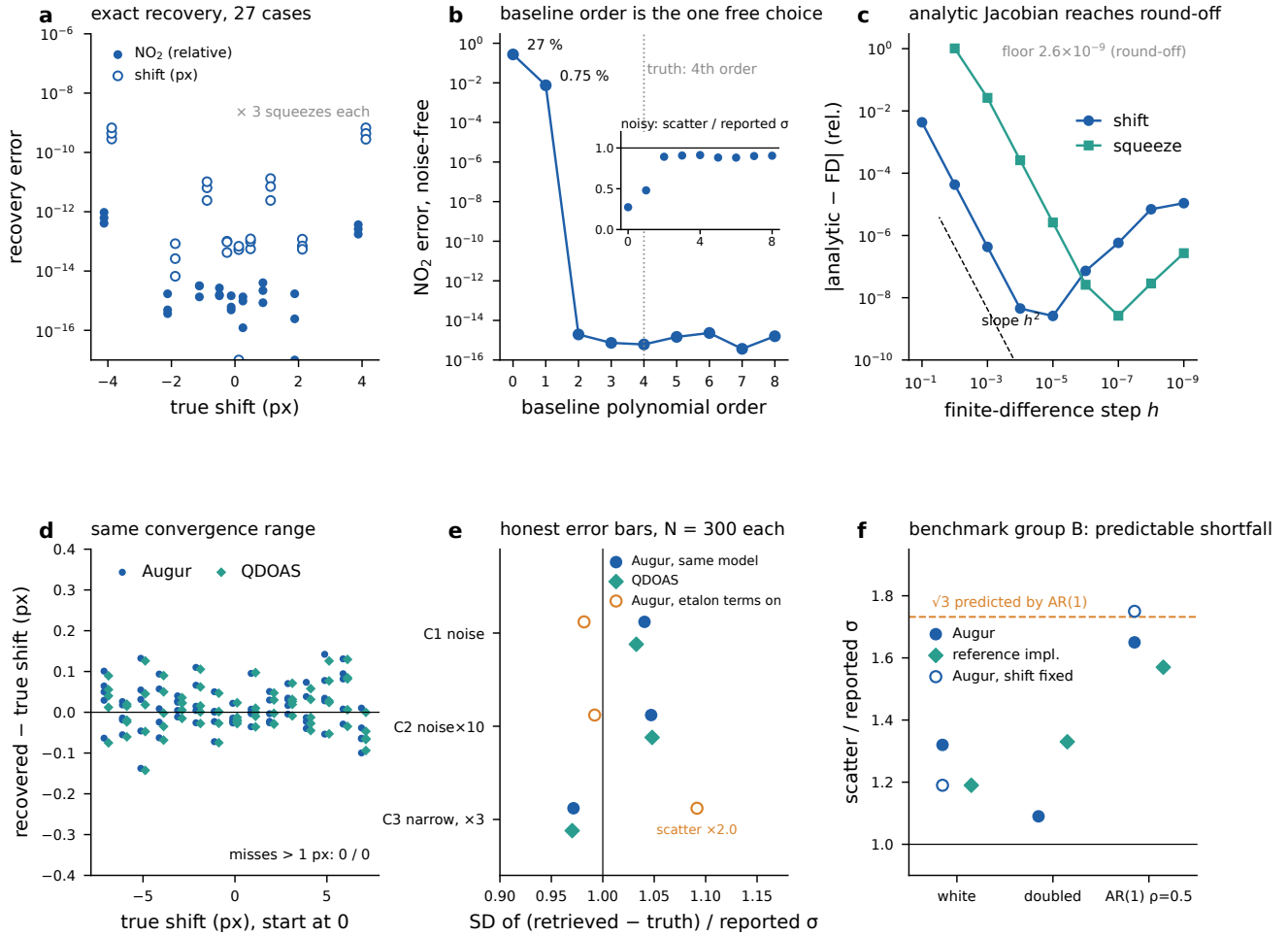


Figure 3. Validation of the retrieval core. **(a)** Error in recovered NO₂ (filled, relative) and shift (open, px) for 27 noise-free spectra (shifts -4 to +4 px; squeezes 0.998, 1.000, 1.002). **(b)** NO₂ error against baseline order for a spectrum built with a fourth-order baseline; inset, realised scatter over reported 1σ on noisy realisations. **(c)** Analytic minus finite-difference Jacobian against step (dashed: slope h^2). **(d)** Shift recovered by Augur and QDOAS minus truth, for true shifts -7 to +7 px (five realisations each, started at 0). **(e)** SD of (retrieved - true)/reported σ over 300 realisations in three noise conditions. **(f)** Benchmark group B: realised scatter over reported uncertainty under white, doubled and AR(1) ($\rho = 0.5$) noise; dashed, the predicted $\sqrt{3}$.

185 be read (Fig. 3a, b). Across the 13 synthetic instruments of Fig. 1a the white-noise ratio ranges from 0.91 to 1.32, within 10 % in 15 of the 30 cases. At nominal noise it lies between 1.22 and 1.32 in 9 of them, where the free wavelength shift adds scatter that the covariance does not carry (Sect. 3.4); under autocorrelated noise it is 1.42 to 2.05.

Table 2. Difference between the analytic Golub–Pereyra Jacobian and a central finite difference of the projected residual, evaluated inside a real fit, against the finite-difference step h , for the shift (px) and the squeeze. Bold: the minimum of each column. The squeeze is a multiplier near unity in a box of order ± 0.02 , so at $h = 10^{-2}$ the step is half the box and the entry 1.02 is a bound artefact; a combined maximum over both parameters would show 1.02 there and hide the V.

step h	shift (px)	squeeze
10^{-1}	4.31×10^{-3}	outside the box
10^{-2}	4.29×10^{-5}	1.02
10^{-3}	4.29×10^{-7}	2.63×10^{-2}
10^{-4}	4.51×10^{-9}	2.61×10^{-4}
10^{-5}	2.60×10^{-9}	2.61×10^{-6}
10^{-6}	7.28×10^{-8}	2.61×10^{-8}
10^{-7}	5.79×10^{-7}	2.64×10^{-9}
10^{-8}	6.95×10^{-6}	2.87×10^{-8}
10^{-9}	1.09×10^{-5}	2.70×10^{-7}

3.2 The analytic Jacobian

The two-term Golub–Pereyra derivative of Sect. 2.3 agrees with a central finite difference of the projected residual down to the round-off floor. The comparison is made inside a real fit: the optimiser call is intercepted and the analytic Jacobian it was handed is differenced against the residual it was minimising. An invariant check of this form runs on every fit in the test suite at the solver’s default step; the step-size sweep of Table 2 turns that check into a measurement.

Both parameters give the expected V (Fig. 3c). The truncation branch falls with a fitted slope of $+2.001$ in each, the h^2 of a central difference, and the round-off branch rises again as the difference of two nearly equal numbers loses significance. The minima are 2.60×10^{-9} for the shift and 2.64×10^{-9} for the squeeze, a ratio of 1.018. A floor common to both parameters is set by round-off in the shared residual evaluation rather than by an error in either derivative. An expression that omitted a term, such as the one-term approximation of Kaufman (1975), would agree only down to the size of the omitted term and then flatten; the descent to the floor identifies the expression as the derivative itself.

3.3 An independent implementation of the same problem

QDOAS (Danckaert et al., 2017) solves the same separable least-squares problem by Marquardt–Levenberg with a singular-value decomposition and a numerically differenced Jacobian. Because it does not model cavities, the extinction spectrum is presented to it as a virtual transmission, $I = \exp(-K\alpha)$ with $I_0 = 1$, so that the optical depth QDOAS forms is exactly $K\alpha$ and both codes solve the same equations on the same numbers.

On synthetic spectra the two codes agree to within the reported uncertainty. On noise-free spectra with shifts of 0, -1.96 , $+2$ and -4 px, Augur recovers NO_2 to 9.5×10^{-13} relative and QDOAS to 5.0×10^{-3} , the latter limited by its output resolution;

Table 3. Agreement of QDOAS with Augur on the field record, per retrieval configuration and species: number of records n in the NO_2 -conditional χ^2 envelope, Deming slope QDOAS/Augur ($\lambda = 1$) and r^2 .

configuration	species	n	Deming slope	r^2
cold	NO_2	36 241	1.039	0.990
cold	CHOCHO	36 241	0.807	0.889
cold	H_2O	36 241	1.377	0.887
hot PNs (180 °C)	NO_2	73 916	1.063	1.000
hot PNs (180 °C)	CHOCHO	73 916	1.195	0.914
hot PNs (180 °C)	H_2O	73 916	1.042	0.974
hot ANs (300 °C)	NO_2	74 539	1.043	1.000
hot ANs (300 °C)	CHOCHO	74 539	1.054	0.959
hot ANs (300 °C)	H_2O	74 539	1.038	0.981

QDOAS recovers the shift to within 0.0042 px. On 30 noisy realisations the two codes have root-mean-square relative errors of 0.0042 and 0.0045 against the same truth, and differ from each other by at most 5.7×10^{-3} relative in NO_2 . In a larger test with 300 realisations in each of three noise conditions (measurement noise, ten times that, and a narrow window with three times the noise), the standard deviation of the error normalised by the reported uncertainty is 0.97 to 1.05 for both codes when they fit the same model. Started from zero, both recover true shifts between -7 and $+7$ px with no miss larger than a pixel (Fig. 3d, e). Switching on Augur’s etalon terms in the narrow window doubles the scatter there, which is a choice of model rather than a property of the solver.

On the field record the two codes agree closely in NO_2 but differ in scale. Over the 53 days, with each spectrum’s measured temperature and pressure used for the unit conversion and records selected by an NO_2 -conditional χ^2 envelope, the NO_2 amounts agree with $r^2 \geq 0.990$ in all three configurations (Table 3). The Deming slope QDOAS/Augur is 1.039 in the cold configuration and 1.063 and 1.043 in the 180 and 300 °C configurations, where $r^2 = 1.000$, so the heated-channel NO_2 scale differs by 4–6 %. The difference is partly tied to the wavelength shift. Within each day the QDOAS/Augur ratio rises with the fitted shift by 0.9 % per pixel in the 300 °C configuration (positive on 49 of 50 days; median fitted shift -5.8 px, both codes finding the same shift) and by 0.6 % per pixel at 180 °C (45 of 49 days). It does not depend on the fitted stretch (Spearman $\rho = -0.006$). Augur shifts the pixel-domain reference, QDOAS the wavelength-domain one through the calibration polynomial (Beirle et al., 2013). The ambient-temperature configuration, whose shift is fixed, still differs by 3.9 %, so a common component of about 4 % remains unidentified. The weak absorbers agree less well (CHOCHO r^2 0.889–0.959, slope 0.81–1.19; H_2O slope 1.04–1.38). Where the codes diverge most – the wavelength shift in the cold window, and with it the weak absorbers there (Sect. 5.1) – the data do not determine the shift, so neither code’s value is established by its fit.

225 3.4 A benchmark that others can run

To make these measurements repeatable with other codes and cross sections we release a synthetic benchmark of 261 cases in six groups. They test exact recovery with no noise (A); reported uncertainty against realised scatter under white, doubled and autocorrelated noise (B); bias from a deliberately incomplete model (C); cases whose wavelength shift is not determined by the data (D); the same truth in two fitting windows (E); and the cost of the sub-pixel interpolator against band-limited truth (F).
230 Truth is known by construction; the spectra are built from the instrument’s own cross sections and wavelength calibration but contain no retrieval code, and the generator is deterministic under a stated seed.

Group B shows that the reported uncertainty can fail for reasons that are calculable in advance. Augur returns scatter-to-reported ratios of 1.32, 1.09 and 1.65 for white, doubled and autocorrelated noise, against 1.19, 1.33 and 1.57 from the reference implementation: both codes pass one condition of three, and they fail different ones. Fixing the wavelength shift
235 at its true value brings Augur’s white-noise ratio to 1.19, so the difference between the codes lies in the nonlinear degree of freedom and its search box, not in the propagated covariance. An AR(1) residual with $\rho = 0.5$ inflates the variance of a sum by $(1 + \rho)/(1 - \rho) = 3$, so a covariance built on independence understates the scatter by $\sqrt{3} = 1.73$. The measured 1.57 and 1.65 sit just below it, and fixing the shift moves Augur to 1.75, because the free shift had absorbed part of the correlated structure (Fig. 3f). A sandwich covariance built from the residual autocorrelation, $\mathbf{G}\Sigma_r\mathbf{G}^\top$ with $\mathbf{G}$ the pseudo-inverse of the design
240 matrix and Σ_r Toeplitz in the residual autocovariance up to its first non-positive lag, brings the autocorrelated case to 1.04 (1.08 with the shift fixed) and leaves the white-noise ratios at 1.31 and 1.08. The joint covariance of Eq. (8) changes none of the ratios by more than 0.1 %.

A deliberately plain profiled grid search is included as a reference implementation whose scorecard shows how a submission is read. The benchmark does not rank codes: groups C and E are reported rather than passed or failed, because no implemen-
245 tation is expected to cover what they measure.

3.5 Relative scale between the two hot channels

The alkyl nitrate product is a difference between two channels, so their relative scale enters it directly. NO₂ was injected into both heated channels, with both ovens at their operating temperatures, at two concentration steps on 2026-08-10/11. The steps did not equilibrate (the delivery line was still passivating), so only the ratio between the channels is used. In the original
250 processing the median of the per-record ratios was 1.228 at both steps. The median channel readings were 14.75 and 11.97 ppb, and 71.57 and 58.16 ppb, whose ratios are 1.232 and 1.231. An origin regression over the 58 one-minute means of one file gives a slope of 0.817 with r^2 0.9998, the reciprocal of the same laboratory ratio. Reprocessed with the current code, block 2053 reads 1.239 times block 4101 (interquartile range 1.231–1.255, $n = 41$ records with block 4101 above 20 ppb).

The relative scale is therefore constant over the concentration steps within this test. It is not constant between calibration
255 states (Table 4): the ratio differs by up to 30 % between them, and the campaign product uses the value of each field state, $g' = 0.952$ for 18 May–5 June, not the laboratory one. The test also leaves the absolute recovery of either channel open and

Table 4. Relative response of the two heated channels, block 2053 over block 4101 (300 °C over 180 °C), by calibration state; field periods in local time (KST).

state	period or date	ratio
field injection, S1	18 May–5 June	0.952
field injection, S2	5–21 June	0.962
field injection, S3	21 June–10 July	1.014
laboratory, current code	August	1.239 (IQR 1.231–1.255)
laboratory, original processing	August	1.228–1.232

cannot separate optical from thermal causes of the offset between the channels, because both cells were at their operating temperatures throughout.

3.6 What is not verified here

260 A co-located reference analyser bounds what this paper can claim. Against the NIER air-quality-monitoring NO₂ analyser at the site (molybdenum-converter type), the operational NO₂ of the heated channels has ordinary-least-squares gains of 0.57 (180 °C) and 0.65 (300 °C), while the ambient-temperature channel has 0.92–1.00. The median ratio of heated to ambient-temperature NO₂ is 0.72, although the heated channels measure NO₂ plus thermally produced NO₂ and should read at least as high. A laboratory injection with the ovens at 300 and 180 °C gave a block-2053 sensitivity of at least 1.08 relative to the
265 cylinder, so oven temperature alone does not explain the discrepancy.

The discrepancy is not a time-base artefact. Before 29 May the heated channels correlate with the analyser at zero lag ($r = 0.985$) and not at ± 9 h ($r \leq 0.03$), and their origin slopes are 0.61 (300 °C) and 0.55 (180 °C) before 29 May and 0.58 and 0.49 after. It is present from the first full day (0.59 on 20 May) and in every calibration state (median ratio to the analyser 0.59, 0.55 and 0.50 in S1–S3), declining somewhat later in the campaign, so it is not caused by a single intervention. Ozone-driven
270 NO₂ loss in the ovens, one of the known thermal-dissociation inlet interferences (Thieser et al., 2016; Dewald et al., 2021; Wüst et al., 2025), is not supported: at night the 300 °C/ambient-temperature ratio rises with O₃ across O₃ terciles (0.69, 0.71, 0.78), and the 180 °C channel reads lower than the 300 °C one. Factors common to the laboratory test and field operation – the $R_L = 1$ assumption, the gas temperature recorded with each spectrum (median 31–33 °C), the cross-section temperature and the retrieval code, which QDOAS reproduces to 4–6 % – cannot produce a factor of 0.6, because the same processing gave at
275 least 1.08 (300 °C) and 0.87 (180 °C) in the laboratory. The laboratory standard entered through the inlet and passed the ovens, so the sampling path as operated there is excluded as well. The remaining candidates are what differed between the two: the field calibration references (reflectivity, helium and zero air), any plumbing used only at the site, the ambient matrix, and the reference analyser, although the ambient-temperature agreement of 0.92–1.00 bounds a molybdenum-converter interference to a few per cent. Separating them needs a standard added at the inlet during field operation and evaluated with the field calibration
280 references, and the cause remains unresolved. The error is multiplicative and leaves the residual, the reported uncertainty and

every context field proposed in Sect. 6 unchanged (Sect. 6.5). No absolute accuracy is claimed. Sections 3.1 and 3.3 verify the retrieval against its own forward model and against another implementation of that model, which bounds implementation error but not model error; model error is what the benchmark of Sect. 3.4 measures.

4 Structural error budget

285 4.1 What the fit reports, and what it cannot report

The covariance returned by the fit (Sect. 2.5) propagates only the residual scatter of the spectrum being fitted. For the operational configuration it gives a single-scan reported uncertainty of 0.55 % on the channel NO₂ amount, the number that appears in the data product. Three inputs of Eq. (1) are not measured with the spectrum but interpolated to its time from neighbouring calibration blocks: the zero-air reference I_0 , the mirror reflectivity $R(\lambda, t)$, and the etalon frequency f , which is held fixed rather than redetermined per scan. An error in any of these enters α directly and, within a single fit, is indistinguishable from a change in the gas amounts; the residual does not see it, so the reported uncertainty does not contain it.

Of two further inputs, sample temperature and pressure contribute negligibly and the instrument line shape could not be tested adequately. Temperature and pressure, interpolated the same way as I_0 , contribute below 0.1 %. The kernel is resampled onto the detector grid with a radius of $\lfloor 4\sigma \rfloor + 1$ pixels, and at the mismatch measured in the ambient-temperature channel (FWHM 0.765 against 0.785 nm) the retrieved amount moved by $\sim 10^{-5}$ %. This shows only that the test did not carry a width change of that size into the forward model; a test with a continuously resampled kernel has not been made, and no mismatch was measured for the heated channels. The residual does carry a signal-proportional model error: the median reduced χ^2 of the 300 °C channel rises from 1.05 at NO₂(180 °C) below 1 ppb to 1.57 at 10–20 ppb. That error is not in the budget below; the cross sections and the line shape are candidates.

300 4.2 Measuring a structural term

A structural term is measured by perturbing an input the way the operational pipeline can actually be wrong, re-running the full retrieval and reading the change in the retrieved amount. In leave-one-out (LOO) perturbation, a jackknife-type resampling of the calibration record (Efron, 1979), one calibration knot is removed, the remaining knots are re-interpolated across the gap, and every scan in it is re-retrieved. Because the interpolated interval is roughly doubled relative to operation, LOO is an upper bound by construction for the interpolation terms (scope in Sect. 4.4). In the split-half estimate, each calibration block is split into contiguous (first and second half) or interleaved (even and odd scans) halves, and the difference of the two block means measures the knot noise with no change of interval. With $n/2$ scans per half the difference carries a factor of two relative to the full-block knot noise, which is divided out.

The ratio of the two estimates separates knot noise from curvature (Table 5). A ratio near unity means the LOO error is entirely knot noise and a shorter reference interval would not improve it; a ratio near one half means the interval carries curvature. For the 180 °C channel the ratio is 0.74–0.91, so measuring the references more often, as recommended qualitatively

Table 5. Zero-air reference-interpolation error in the channel NO₂ amount (per cent) from the split-half estimate of knot noise, with contiguous and interleaved halves, and from leave-one-out (LOO) knot removal, and the ratio split-half/LOO. Hot channels: the 157 zero-air knots of 18–24 May, current fit settings and clock-aligned reflectivity.

channel	split-half, contiguous	split-half, interleaved	LOO	ratio
hot PNs	1.437 %	1.782 %	1.95 %	0.74–0.91
hot ANs	0.90 %	0.97 %	2.25 %	0.40–0.43

by Washenfelder et al. (2008), would reduce the error little, and averaging more scans per block is the lever that could. For the 300 °C channel (0.40–0.43) the interval carries curvature, and shorter intervals would help. The ordering agrees with the direct measurement of Sect. 6.2: tripling the reflectivity interval from 2 to 6 h raises the interpolation error by 15 % in hot PNs and 33 % in hot ANs. Contiguous and interleaved halves give similar knot noise in the NO₂ amount (the contiguous estimate is 7–19 % smaller), so the interleaved split does not understate it.

4.3 Terms must be propagated jointly, and the scale taken once

Measuring each term separately and adding the results in quadrature fails here for two different reasons. First, model-space and α -space perturbations are not additive. Perturbing I_0 and perturbing R both move α , and their effects add scan by scan: the median of $|\delta_{\text{joint}} - (\delta_{I_0} + \delta_R)|$ is 0.8–2.2 % of the signal. The etalon’s sine and cosine columns, in contrast, are free linear parameters that absorb part of any smooth perturbation of α before it reaches the gas columns. Perturbing all three simultaneously therefore gives a smaller result than summing the three responses, by 8.3 % (hot ANs), 6.4 % (hot PNs) and –0.2 % (cold). Adding the etalon term as if it were independent overestimates by an amount that depends on the fitting window and polynomial order.

Second, the joint scale differs from any quadrature combination of the single-term scales, whether robust scales or standard deviations are combined. Combining median-based scales in quadrature misses the directly measured joint scale by +13.3 % (hot ANs), –8.1 % (hot PNs) and +6.9 % (Σ ANs). The same combination on standard deviations does not repair it: the joint exceeds the quadrature sum by 15.5 % for the ANs channel and 7.6 % for Σ ANs, and falls 2.5 % below it for PNs. For variances the failure has two measured parts. The single-term responses of Σ ANs are correlated (pairwise $r = 0.25, 0.18$ and 0.03), which adds 28.7 % to the sum of their variances, and the terms do not superpose (the etalon interaction above), which removes 12.9 %, leaving the joint variance 15.8 % above the sum. Robust scales fail in addition through distribution shape. The normalised shape statistic $s = 1.4826 \text{ MAD/SD}$ is unity for a Gaussian and 0.727 for a Laplace. The three Σ ANs components have $s = 0.73$ (I_0), 0.32 (R) and 0.98 (etalon), and the joint series has a fourth shape again ($s = 0.56$), so a median-based scale captures a different fraction of the variance in each.

Because the sign and size of the discrepancy depend on covariance, interaction and shape that are known only after the joint propagation, no correction factor can replace it. The rule used throughout this paper is therefore to propagate every term

Table 6. Structural error budget, ppb, from a single propagation over the 60 s ambient records of 2026-05-18..05-24 on the clock-aligned reflectivity reference (Sect. 5.4): of the 9034 records with a reflectivity knot within 1 h, the 187 in which a fit of either channel terminated at a bound are excluded (Sect. 5.2), leaving 8847. Σ ANs is formed as in the product, 300 °C minus g' times 180 °C with $g' = 0.9524$ for this period. Columns I_0 , R , f : single-term responses; joint: simultaneous response, the quantity to be used. rSD: robust scale (1.4826 MAD); SD: standard deviation; s : shape statistic of Sect. 4.3.

		I_0	R	f	joint
hot ANs channel	rSD	0.0437	0.0350	0.0082	0.0641
	SD	0.0600	0.0829	0.0086	0.1186
	s	0.73	0.42	0.95	0.54
hot PNs channel	rSD	0.0416	0.0126	0.0187	0.0435
	SD	0.0428	0.0384	0.0189	0.0590
	s	0.97	0.33	0.99	0.74
Σ ANs	rSD	0.0558	0.0269	0.0191	0.0693
	SD	0.0768	0.0831	0.0195	0.1236
	s	0.73	0.32	0.98	0.56

simultaneously into the final product time series, and to take the scale once, at the end, with the same estimator used for the quantity it will be compared against.

4.4 The budget

340 Table 6 gives the budget as the robust scale and standard deviation of the jointly propagated response, in mixing-ratio units. The Σ ANs row is propagated through the channel difference rather than formed from the channel rows (Sect. 4.3). The record evaluated is the product reprocessed on the clock-aligned reflectivity reference (Sect. 5.4), not the dataset delivered to the campaign, which was produced before that correction and is retained as delivered. Records whose perturbed fit terminated at a parameter bound are excluded, because such a fit has not converged (Sect. 5.2). The cold channel was not re-run on this basis
345 and is not included.

The reflectivity term dominates the tail of the budget and the zero-air term its typical record. Of the single-term quadrature sum for Σ ANs, I_0 carries 74 % on the robust scale but 45 % in standard deviation, while R carries 17 % and 52 %. The reflectivity term is small for a typical record and the largest contributor overall, because its error is concentrated in a few intervals: five of the 155 hours in the window carry 57 % of its variance (Fig. 4a, c). Section 6 uses this contrast to choose
350 between remedies.

The budget describes the instrument when its calibration is working. A perturbation experiment can only be run on scans that have neighbouring knots of both reference types, so periods in which the calibration itself failed, which are the periods with

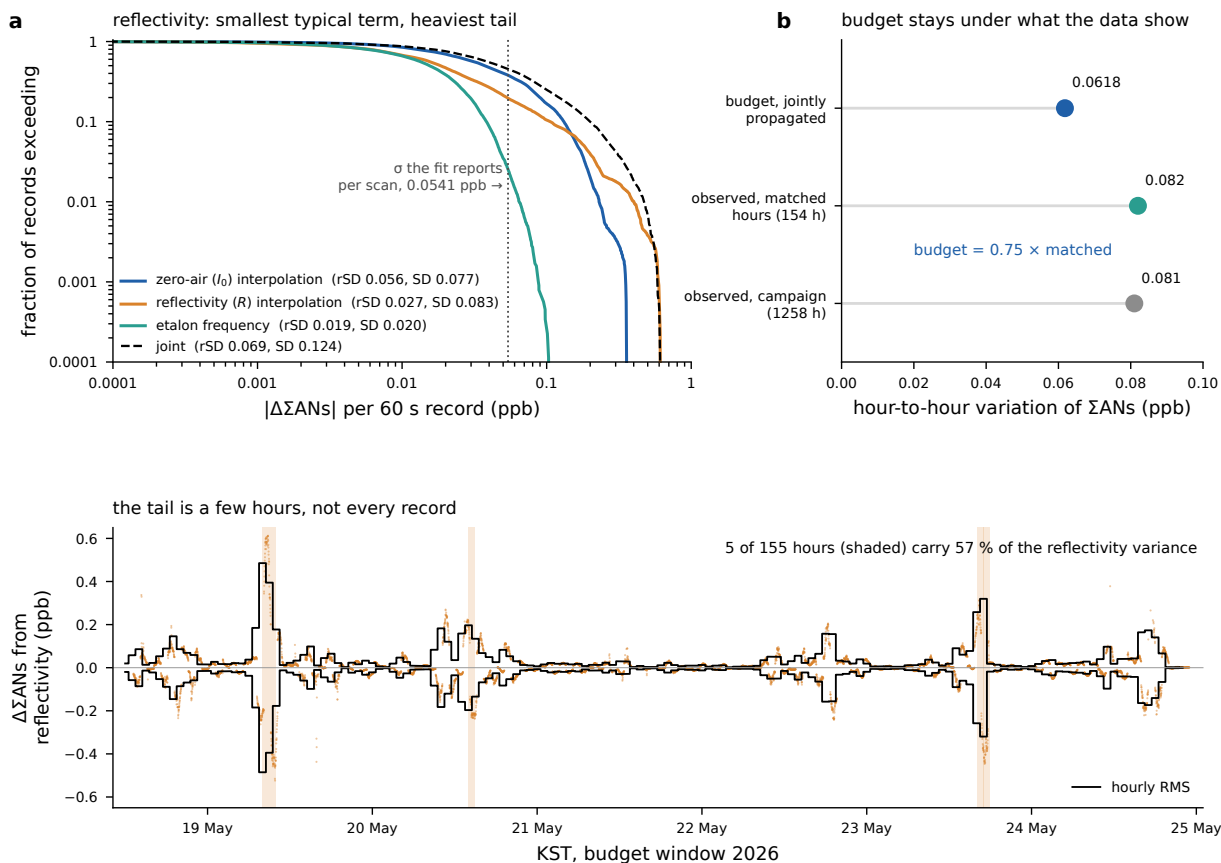


Figure 4. Structural error budget of the alkyl nitrate product over the 8847 records of 18–24 May ($g' = 0.9524$). **(a)** Fraction of records whose ΣANs changes by more than a given amount for each reference term alone and all three jointly; legend: robust scale and standard deviation (Table 6); dotted: the per-scan fit uncertainty. **(b)** Hour-to-hour variation predicted by the joint budget and observed in the matched hours and over the campaign. **(c)** Reflectivity term per record (points) and hourly RMS (steps); shaded, the five hours that carry 57 % of its variance.

the largest errors, are absent from it. The budget window is the best-covered stretch of the deployment, with no record outside the knot range and no gap longer than two hours; over the campaign 11.03 % of cold records lie in such gaps (Sect. 5.3).

355 Against what the fit reports, the budget is large. A budget record is one 60 s ambient bin, which at the operational scan rate is one scan, and the median fit-propagated ΣANs uncertainty at that resolution, over the same records, is 0.0541 ppb. The structural contribution alone is therefore 1.28 times the reported sigma on the robust scale and 2.29 times it in standard deviation. The total uncertainty is 1.63 to 2.50 times what the fit reports, the range spanning the two scales rather than any doubt about the measurement. Because leave-one-out doubles the interpolation interval (Sect. 4.2), these are upper-bound estimates

360 for the interpolation terms in the best-calibrated week of the deployment. They exclude terms common to all knots (cross

Table 7. Hour-to-hour variation of hourly Σ ANs, summarised by $1.4826 \text{median}|\Delta|/\sqrt{2}$ (ppb), observed and predicted by the budget, and the ratio of the budget to the observed statistic in the matched hours. Observed: the clock-aligned Σ ANs product (quality flag below 2), hourly means of at least 8 records.

	gate statistic (ppb)	ratio
observed, campaign (1258 h)	0.081	–
observed, hours matched to the budget (154 h)	0.082	–
budget, hourly, jointly propagated, matched hours	0.0618	0.75
budget, at record resolution	0.0693	0.84

sections, Rayleigh scattering, the reflectivity scale) and periods of failed calibration, and are not a campaign-wide bound. The per-scan fit uncertainty is the appropriate denominator; the uncertainty column shipped with the product, the five-minute fit uncertainty and the zero-air scatter answer different questions (Appendix D, Table D1).

4.5 Consistency with the observed variation

365 The budget can be checked against the data it claims to describe. A channel-differential term must appear in the channel-difference product, and a term that varies on the timescale of the calibration interval must appear in the hour-to-hour variation of that product. The observed variation is an upper bound on the measurement error, because it also contains atmospheric variability. Both sides pass through the same estimator, $1.4826 \text{median}|\Delta|/\sqrt{2}$ over adjacent hourly means, and the budget series is itself hourly (median spacing 1.00 h, p_{90} 1.03 h), so the comparison is made on the same timestamps, paired (Table 7).
370 The lag-1 autocorrelation is -0.60 for the budget, because removing a knot pins the interpolant at the two neighbours and forces it to depart on alternate sides, and $+0.73$ for the observation in the matched hours, from atmospheric persistence.

The budget predicts 0.75 times the observed hour-to-hour variation (Fig. 4b), evaluated on the same 154 matched hours and 153 consecutive pairs against 0.082 ppb; the campaign-wide denominator gives almost the same ratio, 0.76. The budget is not larger than the observed variation, as it must be, nor so far below it as to imply an undescribed dominant error; we assign
375 no pass or fail. The test assumes the error is white at one-hour lag, so any slower structural component is invisible to it; it cannot separate measurement error from atmospheric variability; and the upper bound doubles the interpolation interval by construction. The choice of operator alone moves the ratio from 0.75 to 0.84 (Table 7), which sets the resolution of the method.

Correcting the observed increment statistic for its own lag-1 autocorrelation would defeat the test. Since $\text{Var}(\Delta) = 2\sigma^2(1 - \rho)$ is an identity when ρ is the lag-1 autocorrelation of the same series, the correction returns the standard deviation of the
380 observed series itself, 0.366 ppb for the hourly Σ ANs, which is the full geophysical variability that the differencing was introduced to remove.

4.6 Relation to previous uncertainty treatments

Broadband cavity-enhanced instruments usually report a precision from the spectral fit or from repeated measurements, and an accuracy assembled in quadrature from campaign-constant terms. Washenfelder et al. (2008) derive the mirror reflectivity from the Rayleigh scattering of helium and zero air and bound the absolute accuracy by the uncertainties of the Rayleigh and absorption cross sections, pressure and temperature, summed in quadrature to a slope uncertainty of $\pm 4.6\%$ for NO_2 . Min et al. (2016) use the same scheme, with an NO_2 accuracy of 5.0% limited mainly by the absorption cross sections. Nam et al. (2022) give 2.2% for the reflectivity estimate, dominated by the 2% uncertainty of the zero-air Rayleigh cross section, and 5.2% for the effective path length. In each case the reflectivity enters as a scale factor with a fixed relative uncertainty. The structural budget measures a different quantity: the record-to-record error produced by interpolating intermittently measured references, which is neither a constant scale nor visible in the residual. It adds to the cross-section and Rayleigh terms, which remain multiplicative and are not re-estimated here.

Residual-based error estimates address a third quantity. Platt and Stutz (2008) set out the conditions under which the least-squares error holds, and corrections such as the correlation factor of Stutz and Platt (1996), which accounts for random residual structures and the wavelength-mapping correction, and the bootstrap and Monte Carlo estimates of Hausmann et al. (1999) address structure that appears in the residual. The structural terms here leave the residual unchanged (Sect. 4.1), so an inflation derived from the residual should not recover them. We tested this with the sandwich covariance of Sect. 3.4, which removes the correlated-noise deficit of the benchmark. On the budget records the residual is nearly white (median lag-one autocorrelation 0.06 at 300°C and 0.01 at 180°C); the correction raises the median ΣANs sigma by 12% , and the structural terms remain 1.14 (robust scale) to 2.03 (standard deviation) times the corrected sigma, against 1.28 and 2.28 before the correction on the same records. A further step, not attempted here, would treat the zero-air reference and the reflectivity as measured quantities with their own covariance and propagate the structural term inside the fit rather than afterwards. What the product should carry in consequence is specified once, in Sect. 6.1.

5 What the pipeline already knows and does not report

Section 4 measured errors that leave no trace in the residual. This section concerns the second category: information that already exists in the pipeline – in the profiled objective, the optimiser’s termination state, the knot table and the timestamps of the files being read – and is discarded before the product is written. Reporting it requires neither a better algorithm nor an additional measurement.

5.1 A nonlinear parameter the data do not determine

A converged nonlinear fit has established only that it sits at a stationary point of the profiled residual. If the profiled residual is nearly flat over the range the parameter may take, the returned value is set by the starting point and the bounds, and the uncertainty computed from the local curvature describes a curvature that is not there. Standard DOAS software reports the

parameter value without reporting how much the data constrain it. We test this on the wavelength shift, the parameter least distinguishable from a change in the background. The shift is scanned across ± 4 px with the linear parameters profiled out at
415 each point, a profile-likelihood analysis of practical identifiability in the sense of Raue et al. (2009).

In the cold channel the objective is flat. For twelve ambient spectra the ratio of the largest to the smallest profiled residual sum of squares has a median of 1.0120 and never exceeds 1.0611. In a larger set of 165 spectra the minimum lies at an edge of the scanned range in 77 % of cases, and evaluating the objective at zero shift rather than at its minimum costs 0.01 % to 4.35 %. The same scan in the hot PN_s channel spans a factor of 3.09×10^5 . Operationally the cold shift is fixed at -0.5 px,
420 so the following comparison uses a free shift under settings matched to QDOAS, which shifts and stretches the reference in the wavelength domain (Beirle et al., 2013). With the shift free, the value returned for the same spectra is -2.014 px in this work, with 85.5 % of fits terminating at a bound, against -0.088 px from QDOAS with 4.5 % at a bound; tightening the QDOAS convergence threshold to 10^{-12} changes nothing. The objective map, which involves no optimiser, already places the minimum at an edge in 77 % of spectra, so bound termination follows from the flat objective. Both codes converge and report
425 values two pixels apart, because the data do not distinguish them.

The same channel locates the shift when the absorber is strong enough. On 2 June NO₂ from a cylinder standard (nominally $10 \mu\text{mol mol}^{-1}$ in N₂, past its certification date and used here only for its spectrum) was added to zero air in the cold cavity for half an hour. Retrieved against the preceding zero-air spectrum with the operational reference and window, the plateaus at 9.3 and 3.6 ppb place the minimum of the objective at -0.10 and -0.15 px. Evaluating it at -2.0 px instead costs 39 % and
430 7.5 % of the residual, and nothing for the zero-air spectrum itself. The QDOAS value lies within 0.06 px of that minimum and the value returned here does not; the operational fixed value of -0.5 px costs 2.0 % at 9.3 ppb. The ambient spectra lack the column to locate the shift, not a defined shift (Fig. 5b, c).

The freedom in the shift passes into the retrieved amounts. On the same 5364 cold-channel spectra, the regression slopes between Augur and QDOAS are 0.940 (NO₂), 0.858 (CHOCHO) and 1.027 (H₂O) when Augur's shift floats, and 1.019, 0.662
435 and 1.178 when it is fixed at -0.5 px. The weakest absorber thus moves by 23 % according to a parameter the residual cannot resolve. For two codes solving the same problem every species should give unity, and none does in either treatment; with the shift fixed, CHOCHO and water still differ by 34 and 18 %, so the disagreement is not the shift alone. The laboratory spectrum shows that the shift this channel supports is close to zero, -0.10 px. The operational fixed value is 0.4 px from it, and replacing -0.5 by -0.1 px changes the slopes against QDOAS by at most 2.2 % (NO₂ -0.06 %, CHOCHO $+2.2$ %, H₂O -0.8 %); the
440 floated value is 1.9 px from it and moves CHOCHO by 23 %. Whether the shift is fixed, not the value at which it is fixed, divides the results, and neither fit report shows it.

The budget of Sect. 4 is not an artefact of basin hopping in the shift, and the reflectivity term hardly couples to the shift at all (Appendix D, Table D2).

Reporting -2.014 ± 0.03 px for a quantity the data locate only to ± 2 px is the failure described here, and for this channel
445 the resulting term is larger than any in Table 6. The ratio of the objective at the returned value to that at a fixed reference point, which the fit evaluates anyway, would flag it (Sect. 6.1). Because the shift belongs to the spectrograph rather than to the air, it

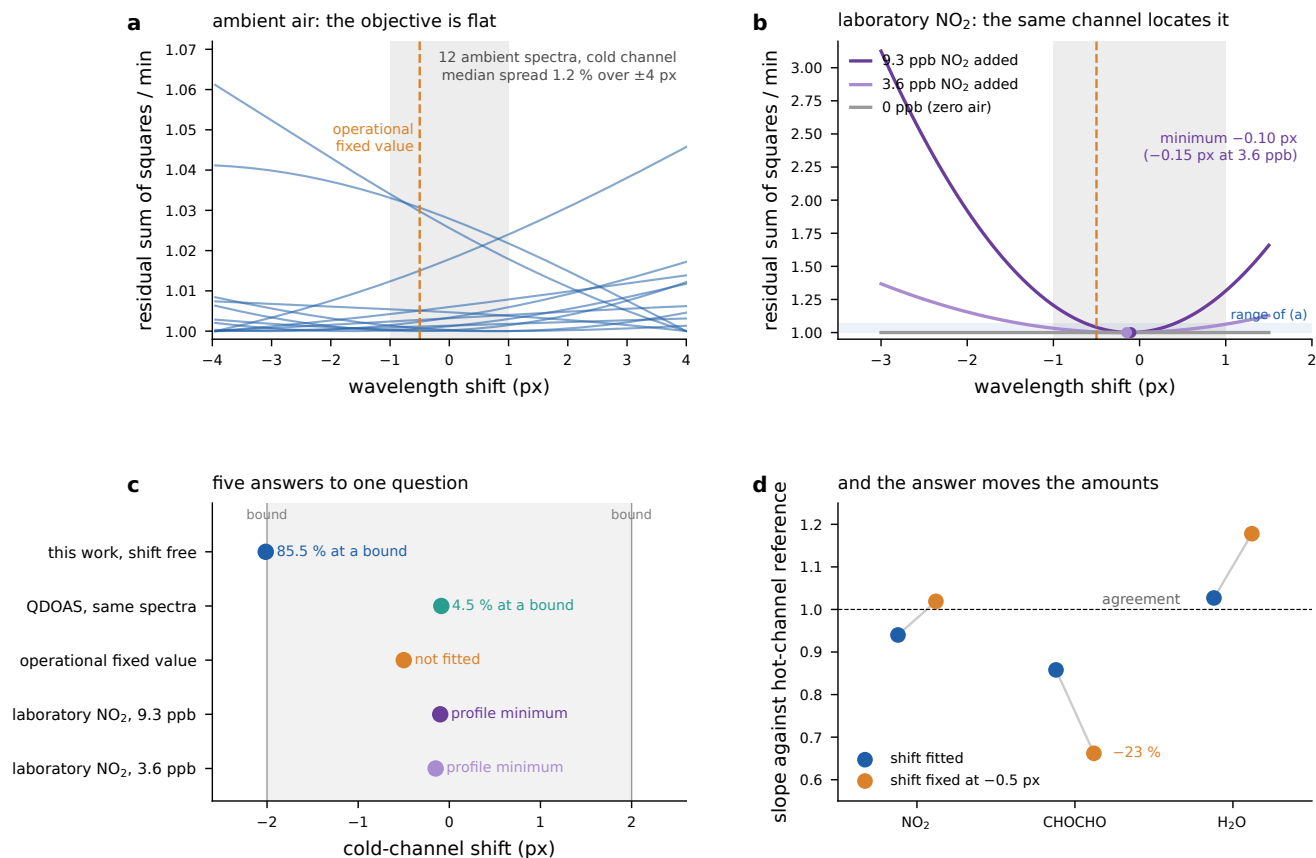


Figure 5. Identifiability of the cold-channel wavelength shift. **(a)** Profiled residual sum of squares against shift for twelve ambient spectra, each normalised to its minimum; shaded, the range the operational configuration allows; dashed, the value it fixes. **(b)** The same scan for the laboratory spectra of 2 June (zero air with 9.3 and 3.6 ppb NO₂ added, and zero air alone); band, the vertical range of (a). **(c)** Shift returned for the ambient spectra by two implementations under matched settings (with the fraction of fits at a bound), the operational fixed value and the profile minima of (b). **(d)** Regression slope between the cold-channel retrievals of Augur and QDOAS on the same spectra ($n = 5364$), with Augur's shift fitted and fixed.

could also be fitted jointly across the spectra of a calibration block, as a separable problem with common nonlinear parameters. This is not implemented here and would need an absorber in the block, as in the laboratory spectrum of Fig. 5b.

5.2 Termination at a bound is a routine outcome

450 In this instrument bound termination is the normal state of one channel and a systematic response to perturbation in another. The operational configuration (Appendix A) bounds the shift to $[-10, +0.5]$ px in hot ANs and $[-5, +5]$ px in hot PNs and fixes it at -0.5 px in the cold channel. In a sensitivity run that instead bounded the cold shift to $[-1, +1]$ px, every one of the 22

Table 8. Calibration coverage of the retrieved records, from calibration blocks extracted directly from the raw files: number of records n ; percentage of records outside the knot range and inside a knot gap longer than 2 h (counts from the coverage extractor; the retrieval itself has 9149 ambient records per heated channel in the budget window), for the zero-air (I_0) and reflectivity (R) references; and the longest reflectivity gap.

instrument	window	n	I_0 out of range	I_0 gap > 2 h	R out of range	R gap > 2 h	worst R gap
hot ch1/ch2	campaign	76 282	0.08 %	0.07 %	3.76 %	1.23 %	10.6 h
hot ch1/ch2	budget window 05-18..24	9 088	0.01 %	0.00 %	0.00 %	0.00 %	1.3 h
cold	campaign	43 259	0.13 %	0.34 %	2.89 %	11.03 %	11.8 h
cold	cold window 06-10..16	7 069	0.42 %	0.00 %	17.32 %	18.31 %	11.8 h

base fits returned exactly -1.000 px, the bound itself, and was reported as converged. In the production propagation of Sect. 4 the cold shift is held fixed, as in operation, so the channel contributes no information about the parameter.

On the clock-aligned basis of Table 6, 187 of the 9034 Σ ANs records are excluded because a fit of either channel, unperturbed or perturbed, terminated at a bound. The split by perturbation was recorded only for the earlier run on the unaligned reference: there, in hot PN_s, the zero-air perturbation accounted for 121 bound-terminated records against 4 for the unperturbed fits (the categories overlap), and the hot AN_s channel had none. In that run the operational fit was well inside its feasible region and the structural perturbation drove it out, so the margin between the operational solution and the edge of identifiability is smaller than the size of a reference error. Augur computes the termination states internally; the delivered campaign product did not carry them, and the current version exports them as separate columns (Sects. 2.6 and 6.1).

5.3 How far a record is from its calibration

The distance in time from a record to the calibration knots that produced its references varies over the campaign by more than an order of magnitude, and the pipeline computes it in order to interpolate and then discards it. The retrieval interpolates the reflectivity across an 11.8 h gap exactly as across a one-hour gap and returns the same uncertainty in both cases. Coverage was measured over the whole campaign from calibration blocks extracted directly from the raw files, independently of the retrieval. The extraction reproduces the production knot cache (157/157 hot and 164/164 cold knots at 0.000000 s) and the alpha header’s own `ZA_count=1313`. Two distinct deficits appear: records outside the knot range, and records inside a knot gap longer than two hours (Table 8).

For the zero-air reference the interpolation is genuine: 0.08 % of hot records lie outside the knot range and 0.07 % inside a gap longer than two hours, with the nearest knot a median 922 s away. The zero-air knot set is on the corrected time axis – the production cache over 05-18..24 reproduces a re-extraction at 0.000000 s – so the I_0 column is unaffected by the displacement of Sect. 5.4.

The cold channel’s deficit is a gap problem. The existing quality gates reject 11.3 % of cold pairs against 0.24 % of hot pairs (3 of 1 264); an independent count gives 75 of 743, 10.1 %, and nothing below turns on the difference. Independent rejection at

Table 9. Change in the retrieved amounts when the heated channels are retrieved with the clock-aligned instead of the nine-hour-displaced reflectivity knots, over the records of the budget window: robust scale (rSD) and standard deviation (SD) of the change (ppb), and the rSD of the change relative to the rSD of the R term in Table 6. Σ ANs is formed with $g' = 0.9524$ as in Table 6.

series	delta rSD (ppb)	delta SD (ppb)	vs the R term of Table 6
ANs channel	0.1048	0.1627	3.0
PNs channel	0.0358	0.0563	2.8
Σ ANs	0.100	0.1418	3.7

the 11.3 % rate would leave 3.5 % of cold records in gaps longer than two hours, against the 11.03 % observed. The worst gap, 11.8 h, is eleven consecutive rejected pairs, a run expected 0.03 times in the campaign even at a 40 % independent rejection rate. The rejections arrive in runs, the episodic structure that Sect. 6.1.1 finds in the reflectivity variance itself. The same failures are absent from the budget population (Sect. 4.4). In the cold channel, eleven of the thirteen zero-air blocks identified as defective in Sect. 6.3 fall outside the case set, because no reflectivity knot exists within 2.8–5.8 h of them. A collapse of the zero-air signal takes the He/zero-air pair down with it, so the two reference failures are one event seen twice. The interpolation already holds the time to the nearest knot, the length of the enclosing interval and whether a record is out of range. Reported, they would separate the 0.00 % of the budget window from the 18.31 % of the cold window, which no field in the current product distinguishes (Sect. 6.1).

5.4 A reference error the fit does not see

The operational reflectivity reference was not clock-aligned. Its knots for 2026-05-18 to 05-30 lie on a time axis that a later correction to the raw-data reader superseded, while the spectra lie on the corrected one. Every record in that span, including the budget window of Sect. 4, was therefore retrieved with a reflectivity displaced by nine hours. The zero-air reference is unaffected, because its knots were already on the corrected axis (verified to 0.000000 s), and so is the cold channel, to which the correction does not apply. The retrieval was run twice over all 9 132 records of the window common to both heated channels, changing nothing but the knot time base (Table 9).

The displacement moves the alkyl nitrate product by 0.100 ppb, between three and four times the reflectivity term of the budget and 1.44 times the whole of it; the ratio compares a systematic displacement with a record-to-record budget and indicates scale only. It is the largest term we can quantify within the retrieval; the multiplicative scale discrepancy of Sect. 3.6 is larger but lies outside it. It is not in Table 6, which measures what a reference’s own uncertainty costs rather than what using the wrong reference costs.

The fit report does not register the displacement (Table 10). The extinction spectrum changes by an amount comparable to the single-scan noise floor of 6.16×10^{-9} , and the residual and the reported uncertainty change by the same two parts in a thousand. The agreement is expected, because the reported uncertainty is a near-constant multiple of the residual RMS (Sect. 7.3); an unchanged residual and an unchanged uncertainty are one witness reported twice. The fitted wavelength shift

Table 10. What the fit report shows for the displacement of Table 9: RMS change of the extinction spectrum (cm^{-1}), ratio of the residual and of the reported σ between the two retrievals, change in the retrieved amount in units of the reported σ , and change in the fitted shift.

	α RMS change	residual ratio	reported σ ratio	$ \delta c /\sigma$	$ \delta \text{shift} $
ANs	3.37×10^{-9}	1.0020	1.0022	1.99	0.0053 px
PNs	1.53×10^{-8}	1.0016	1.0016	0.55	0.0065 px

moves by 0.005 px, where the reference perturbations of Table 6 move it by 0.243 px (median), about forty times more, so the error is neither absorbed by the nonlinear parameters nor revealed by the residual. The retrieved concentration of the 300 °C channel moves by two reported standard deviations while every quality indicator in the fit report stays the same.

Correcting the time base changed the budget in both directions. The robust scale of the Σ ANs budget fell by 7.0 % (0.0745
505 to 0.0693 ppb) while its standard deviation rose by 4.5 % (0.1183 to 0.1236). The total uncertainty relative to what is reported therefore widened at both ends, from 1.70–2.41 to 1.63–2.50 times the same fit-reported sigma (Sect. 4.4). Coverage improved, from 8505 to 9034 of the window’s 9149 ambient records (93.0 to 98.7 %), because the misaligned knots had been falling outside the one-hour window that admits a record to the perturbation experiment. The correlation between the two channels’ joint responses fell from +0.412 to +0.146: because Σ ANs is a channel difference, a common-mode error partly cancels in
510 it, and the pre-correction cancellation was an artefact of an error the two channels shared. Both effects were properties of the contaminated product, and the budget measured on it inherited them.

The cost of the displacement depends on how fast the reflectivity was moving. Across the boundary where the clock was corrected, the affected side gives 0.0622 ppb against 0.0000 ppb on the unaffected side, confirming that the realignment touched only the intended knots; near the boundary the reflectivity moved more slowly than in the budget window. The budget-window
515 figure of 0.100 ppb is the value to quote; 0.0622 ppb is range information, not a worst case.

The underlying cause was a stale derived file. The heated-channel clock itself ran on local time until 29 May and was corrected in the reader after comparison with an external time reference. The reflectivity file was written on 2026-07-10, the correction that invalidated it was committed on 2026-09-19, and the retrieval read both without comparing them. Recording, for each derived reference, the version of the code and the raw files it was built from detects this condition mechanically
520 (Sect. 6.1); it is the only failure in this paper that admits a complete remedy rather than a better report.

6 What should be reported, and what should be changed

This section states what a DOAS product should carry instead of a single uncertainty, and what in the measurement schedule is worth changing. The knot gate (Sect. 6.3) and the coverage fields (Sect. 6.4) are implemented in Augur and evaluated against the campaign. The reporting convention of Sect. 6.1 is implemented in part: the termination state, the distance to the nearest
525 knot and the identity of the reference file are exported, while the build time and time convention of each reference are specified for implementation; Sect. 6.5 names the errors that none of these measures reaches.

Table 11. Fields of the context flag, reported per record and per reference type (zero air, reflectivity), and the reason for each.

field	why
seconds to the nearest knot, with the side (before/after)	the side distinguishes interpolation from extrapolation
length of the bracketing interval	an eleven-hour gap and a fifteen-minute one are not the same interpolation
in-range flag	extrapolation is a different operation and should be nameable in a filter
number of adjacent calibration blocks rejected by the quality gates	the gates’ cost, which Sect. 6.4 argues must be reported wherever the gates are applied
identifier of the reference file, and when it was built	two products made from different calibration vintages are not comparable
the time convention the reference was built under	a correction applied upstream makes every earlier derived product stale

6.1 Three numbers per retrieval, not one

Every retrieved amount should be reported with three numbers:

1. the fit uncertainty, from the covariance of the projected problem (Sect. 2.5), which describes what the residual supports.
- 530 2. the structural term, evaluated for that scan’s own calibration context rather than as a campaign constant (Sect. 4).
3. a context flag recording the state of the references the scan depends on.

The fit uncertainty is a statement about the residual of a model assumed correct, not about the inputs to that model, and keeping the three separate preserves that distinction for whoever uses the product. Reporting per-record uncertainty components with different correlation scales (random, structured, common) is established practice for Earth-observation climate data records (Merchant et al., 2017), building on JCGM (2008); the three-number convention applies it to a single retrieval. The fit un-

535

certainty should travel with the termination state of the nonlinear stage (Sect. 2.6), and, where a nonlinear parameter may be unidentifiable, with the ratio of the objective at the returned value to that at a fixed reference point (Sect. 5.1). The value of the context flag lies in which fields it carries, so we specify them per record and per reference type (zero air, reflectivity) in Table 11.

540 The last two fields address the failure of Sect. 5.4. No provenance field detects the clock error of the raw record itself (Sect. 5.4). The fields catch the failure that followed: a reflectivity reference built before the correction and never rebuilt. Any correction applied upstream invalidates every derived product built before it, and a pipeline whose derived products do not record their build convention cannot tell that this has happened.

545 The mismatch was visible in the data as a nine-hour run of out-of-range records at the start of the deployment – the first zero-air block at 02:42, the first reflectivity knot at 11:40 – with 42.3 % of that day extrapolated. The distance and in-range fields would have carried that anomaly into the product on the first run after the correction, rather than two months later. Identifying its cause needed the build time of the reflectivity file and the convention it was built under, so detection and diagnosis need

Table 12. Concentration in time of each structural term over the hours of the budget window (clock-aligned basis of Table 6): shape statistic s ; number (and fraction) of hours holding 90 % of the term’s variance; the number of contiguous runs those hours form, with the number expected for the same count of hours drawn at random; share of the variance on the standard-deviation basis; and the remedy the shape indicates.

term	s	hours holding 90 % of the variance	contiguous runs (ran- dom expectation)	share of variance (SD)	remedy
I_0	0.73	63 (41 %)	24 (38)	45 %	longer calibration blocks (untested, Sect. 6.2)
R	0.32	26 (17 %)	12 (22)	52 %	detect and flag intervals (Sect. 6.3)
etalon f	0.98	118 (76 %)	26 (29)	3 %	fit f per scan, or model it

separate fields. The same tags allow a load-time check that refuses a reference whose time-convention tag differs from that of the spectra; we recommend it where the tags exist, and the report in every case so that the error stays interpretable.

550 The structural term cannot be obtained in closed form. The amplification factor $I_0/(I_0 - I)$, which governs how an error in the zero-air reference enters the extinction, varies by a factor of forty between records and is available without fitting. Regressed against the measured response, however, it explains between 0.0 % and 9 % of the variance, with the wrong sign. The polynomial background absorbs the smooth part of the perturbation, so what reaches the gas columns is set by shape rather than amplitude. Normalising the response by the retrieved amount does not help: the robust coefficient of variation is 0.80–1.09

555 in relative terms against 0.83–1.11 in absolute. The structural term therefore has to be obtained by running the perturbation, which is available in post-processing but not in a real-time monitor, and the two implementations differ accordingly.

6.1.1 Which remedy a term needs is readable from its shape

The shape statistic s of Sect. 4.3 indicates which of three actions a structural term needs. A term with $s \approx 1$ is Gaussian: its error is spread evenly over all records and can only be reduced by a better model. A term with $s \approx 0.7$ –1 is dominated by

560 measurement noise on the references and responds to averaging. A term with $s \ll 0.7$ is concentrated in a few records and needs detection rather than averaging. This prediction is checked by ordering the 155 hours of the budget window by their contribution to each term’s variance (Table 12).

The ordering by s is the ordering by concentration: the two-hour interval from 23:00 UTC on 18 May to 01:00 UTC on 19 May alone holds 37 % of the reflectivity term’s variance. The reflectivity hours form far fewer contiguous runs than the same

565 number of hours drawn at random, so they are episodes rather than scattered outliers, and a flag can address them. The etalon hours are indistinguishable from random, the signature of a term no flag will help. On the robust scale the ordering of the first two terms reverses (Sect. 4.4), which is why Table 6 reports both scales. A term that is small for a typical record and largest in total is the one a campaign-mean uncertainty hides.

6.2 Calibration blocks: interval length and scans per block

570 The natural response to an interpolation error is to interpolate over shorter intervals. How much that buys depends on the channel. Tripling the reflectivity interval from 2 to 6 h was measured to increase the interpolation error from 0.87 % to 1.00 % in hot PN's and from 2.05 % to 2.73 % in hot AN's. In the 180 °C channel the error is set mostly by the noise on the knots, and a shorter interval would buy little; in the 300 °C channel the interval carries curvature (Table 5), and shorter intervals would help.

575 The other lever is the number of scans averaged into each calibration block (Appendix C). Within the operational block ($N = 34$ scans, $T = 32$ s), doubling the number of scans averaged reduces the scatter of contiguous sub-blocks by 7 % in the AN's channel, against the 29 % that independent scans would give. The measurable range lies below the operational block length, so the evidence is not decisive; operating one longer block would settle it. Even with white averaging the return would be modest. The zero-air term carries 45 % of the Σ AN's variance, so a fourfold block, which would raise the calibration duty from

580 1.20 to 3.89 %, would lower the structural budget from 0.1236 to 0.1008 ppb (18.5 %). Flagging the few defective reflectivity intervals would lower it to 0.0901 ppb. That ordering – flagging before averaging – holds whatever the block scaling turns out to be, because the reflectivity term is concentrated in a few episodes and the zero-air term is not (Sect. 6.1.1).

6.3 Gating defective knots

A few calibration blocks carry most of the zero-air term. Ordering the cold-channel zero-air blocks by their leave-one-out

585 influence, the worst single block carries 28 % of the total variance, the worst three 76 % and the worst five 97 %, so a structural term computed over all blocks is a statement about a handful of them. They are identifiable before any fit is attempted. The block's light level relative to the campaign median, with the threshold $I/\tilde{I} < 0.5$, selects only 13 of the 38 blocks whose influence exceeds 2 %. Those 13, however, include nine of the worst ten and account for 99.9 % of the influence variance; the 25 blocks it misses are all in the 2–3 % range, and the threshold excludes 8 % of blocks. Classification accuracy would have

590 rejected this threshold, but the relevant figure of merit is the variance removed, and a single observable removes essentially all of it.

Only the zero-air reference needed this gate. Reflectivity knots already pass three quality gates: a minimum fraction of valid pixels, an acceptance band on the retrieved effective path length, and limits on the calibration residual. A defective helium or zero-air pair is therefore rejected before it becomes a knot, whereas every zero-air block became a knot regardless of how much

595 light reached the detector. The gate is implemented with the threshold expressed as a fraction of the run's own median block intensity rather than as an absolute count, so it transfers between instruments, and it is disabled by default so that reprocessing an existing product is an explicit choice.

6.4 Coverage: say when the reference is far away

The coverage deficits of Table 8 have two causes that cost different things to remove. Extrapolation (records outside the knot

600 range) and gaps (records between knots more than two hours apart) both concern the reflectivity, since zero air has neither. The

first cause is bookkeeping and the cheapest fix in this paper: reflectivity generation stopped at 07-09, which produces the hot channel's trailing extrapolation and is removed by re-running over files that already exist. The second is the clustered rejection of cold calibration pairs described in Sect. 5.3; its remedy is to restore the light level or, failing that, redundancy. Only this second cause concerns how often the instrument calibrates, and even there the average rate is not the problem, because the
605 rejections arrive in the same episodes that carry the reflectivity variance (Table 12).

Quality control has a coverage cost. In the cold channel the reflectivity gates are, correctly, the proximate cause of the coverage deficit. This qualifies the argument of Sect. 6.3 for a zero-air gate: gates remove variance and coverage, and a pipeline that gates without reporting the cost has moved the error somewhere harder to see. The context flag of Sect. 6.1 records that cost. The same caution applies to the budget. The reflectivity contribution of Sect. 4 is measured by removing a knot and re-
610 interpolating, so its magnitude scales with the knot spacing, and a sparse reference appears as a dominant term partly because it is sparse. The remedy therefore belongs to the schedule and to reprocessing rather than to the retrieval. Records in gaps, which the budget cannot cover, are reported with the context flag set and excluded from any statistic that assumes a characterised uncertainty.

Each reflectivity knot pairs an hourly zero-air block with the most recent accepted helium block, measured every three
615 hours (Appendix A, Table A2). A rejected zero-air/helium pair removes one hourly knot and leaves a two-hour gap; a rejected helium block removes no knot but makes the following knots use helium measured up to six hours earlier. Matching the helium cadence to the zero-air cadence would bring the total calibration duty from 1.20 % to 1.79 % and make the helium half of every knot contemporaneous. Whether it would also shorten the gaps depends on why pairs are rejected, which in the cold channel is mainly the light level. The case for it is redundancy against runs of rejection and contemporaneity, not precision: in the 180 °C
620 channel the reflectivity error depends only weakly on interval length (Sect. 6.2). At 1.2 % of the measurement time, calibration is the binding constraint on none of these remedies.

6.5 Errors that neither the fit report nor the proposed fields detect

The convention of Sect. 6.1 makes visible what the pipeline already knows. It cannot make visible what the pipeline does not know, and two such errors appear in this record.

625 The first is a multiplicative gain error. Against a co-located NIER analyser the heated-channel NO₂ reads low by a factor of about 0.6 while the ambient-temperature channel agrees (Sect. 3.6). A gain leaves the residual, the reported uncertainty, the termination state and every distance and provenance field unchanged, and the zero-air test of Sect. 7.2 is blind to it because a gain applied to zero is zero. Only a comparison with an independent measurement of the same quantity reveals it, and does not by itself identify the cause.

630 The second is the raw light level. All six ambient-temperature excursions of Sect. 7.3 coincide with spectra that fell below 30 % of their usual count at 460 nm; the residual finds them, and the reference context is abnormal in five of them, but none of the proposed fields names the cause. The light level of each spectrum, and the helium and zero-air intensities of each calibration block rather than only their ratio, are known when they are recorded and could travel with the product as further context; we have not evaluated them for the heated channels.

Table 13. Noise floor of the Σ ANs product, measured on the paired zero-air blocks of the two heated channels: 1σ , 3σ detection limit, median at a true zero, and the ratio of the measured 1σ to the uncertainty column shipped with the product. Both scales are computed on the 155 paired blocks with $g' = 0.9524$; the inter-channel correlation is $r = -0.093$.

	robust scale	standard deviation
measured 1σ	0.0733	0.0856 ppb
detection limit, 3σ	0.220	0.257 ppb
median value at a true zero	+0.0053 ppb	+0.0053 ppb
against the product-column 0.0806 ppb	0.91	1.06

635 **7 Field demonstration**

7.1 Instrument and deployment

The field case is a thermal-dissociation broadband cavity-enhanced absorption spectrometer with two heated and one ambient-temperature cavity, of the design described by Min et al. (2016) and Nam et al. (2022). Ambient air is split between a cell held at 180 °C, in which peroxy nitrates dissociate, and one at 300 °C, in which alkyl nitrates dissociate as well (Day et al., 640 2002; Wooldridge et al., 2010); a third cell at ambient temperature measures NO₂ directly. The three differ in what has been converted to NO₂ upstream rather than in how the spectrum is formed, and the sums of peroxy and alkyl nitrates follow as channel differences, Σ ANs with the inter-channel factor g' of Sect. 4.4. The retrieval configuration of each channel – window, baseline, references, shift and squeeze settings, wavelength calibration, line shape, cavity constants and calibration schedule – is given in Appendix A.

645 The instrument was deployed at Suncheon Shindae (SDD) from 2026-05-18 to 07-10 as part of a multi-institution intensive campaign. Two properties belong to this deployment rather than to the configuration. The median fitted shift in the 300 °C channel is −6.12 px, inside its bound of −10 px. The field numbers of this paper were checked against a reprocessing with a corrected pressure-sensor pairing, which changes the channel NO₂ by ±0.25 %, and are reported as first computed. The instrument itself, its inlet and its conversion efficiencies are characterised elsewhere.

650 **7.2 Noise floor of the product record**

The noise floor of a channel-difference product has to be measured on the difference. The two hot channels share a light source, a cavity and a clock, and their zero-air blocks coincide in time, so 155 blocks pair exactly and the difference can be formed record by record (Table 13). The channels turn out to be nearly uncorrelated, $r = -0.093$, so a quadrature assembly would have come within about 3 % of the measured value; on an instrument whose channels shared a drift it would not, and nothing 655 in the fit report distinguishes the two cases.

The median at a true zero is +0.0053 ppb in the difference (+0.0028 and −0.0020 ppb in the channels). A product assembled through zero-air interpolation, a Rayleigh correction, a time-interpolated reflectivity, a spectral fit and a channel subtraction

therefore returns zero when the truth is zero, to a fortieth of its own detection limit. This closes the additive chain end to end; a zero test is blind to gain errors, such as the scale discrepancy of Sect. 3.6.

660 The zero-air floor is roughly covered by the uncertainty column shipped with the product, at 0.91 to 1.06 times that column's 0.0806 ppb, depending on the scale. This does not conflict with Sect. 4: the zero-air floor uses the same references minutes apart, whereas the structural budget measures what happens when a reference is wrong. In 46.1 % of the budget records the structural change exceeds that record's own fit-propagated σ . The product column is 1.48 times the quadrature of the channel columns, 0.0806 ppb against 0.0543, because it is the inter-channel gain systematic of Appendix D, which scales with the NO₂ 665 amount; the ratios of Sect. 4 are therefore taken against the fit-propagated sigma.

7.3 The reporting convention in operation

Applied retrospectively to this deployment, the quantities of Sect. 6.1 separate what the fit report catches from what it does not, in the heated and ambient-temperature channels alike (Fig. 6); the provenance fields among them are specified but not yet implemented. In Fig. 6f the residual flags an episode when the reduced χ^2 exceeds 1.5, and the distance field when the knot 670 interval exceeds 2 h or a knot departs by more than 50 % from its running median. The light level flags it when the ambient or zero-air count falls below 30 % of its usual value; it was not assessed for the heated channels, and the cold configuration, with fixed shift and squeeze, has no bound information in its termination state.

The fit report does not catch the displacement of Sect. 5.4 at all. Across the deployment the reported uncertainty of the ANs channel is a near-constant multiple of the residual RMS ($r = +0.999$ over all 74 667 records; the ratio varies by 2.8 % between 675 its 5th and 95th percentiles), so the two are one witness. Among the proposed fields, the distance and in-range fields flag these records and only the provenance fields would name the cause.

The ambient-temperature channel supplies the opposite case. Its six excursions below -5 ppb (19 hourly medians between 19 May and 2 June) are all visible in the fit itself: the residual RMS is 8 to 600 times its campaign median and the reduced chi-square 5.6 to 91, against 1.2 in ordinary hours. Five of the six coincide with an abnormal reference context: the bracketing 680 reflectivity knots are 2.6 to 9.0 h apart against 1 h nominal, or one of them departs from its running median by 66 to 300 % against a typical 10 %. The distance and knot-validity fields therefore explain them but are not needed to find them. The sixth, four hours on 2 June, has an ordinary reference context and is identified by the residual alone. All six share a proximate cause that none of these fields names, the raw light level. In each, the ambient or the zero-air spectrum, or both, fell below 30 % of its usual count at 460 nm, in four of them to within a few hundred counts of the detector dark level. On 2 June only the 685 ambient path lost light (22 % of the zero-air level), which is why the reference stayed ordinary. The misfit threshold is itself signal-dependent (Sect. 4.1), and these excursions exceed it by an order of magnitude or more.

In the heated channels no hourly median falls below -0.5 ppb over the deployment. The termination-state field flags one episode: for eleven minutes on 21 June the 180 °C channel returned its shift at the $+5$ px bound. The residual is raised over the same records (reduced chi-square 4.7 to 5.5), so the residual signals a problem and the termination state identifies it. For 690 the displacement and the bound episode, one of the context quantities supplies what the fit report lacks; the light level and the heated-channel scale are beyond all of them (Sect. 6.5).

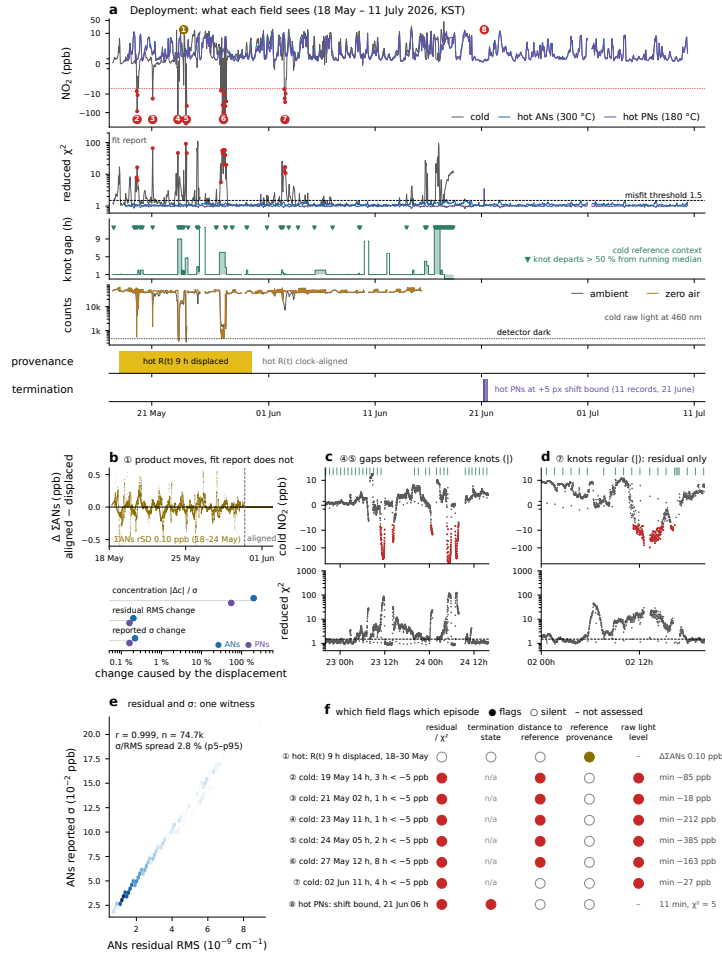


Figure 6. What each reporting field flags during the deployment (18 May – 11 July 2026, KST). **(a)** Hourly median NO₂, three configurations (red: cold hours below −5 ppb); hourly median reduced χ^2 ; cold-channel reflectivity knot interval (triangles: knot departures > 50 %); cold-channel light at 460 nm; heated-channel reflectivity provenance; records at a shift bound. Badges: episodes of **(f)**. **(b)** Episode ①: change in ΣANs between clock-aligned and 9 h displaced reflectivity (line: 6 h median); below, for 18–24 May, in units of reported σ . **(c, d)** Episodes ④, ⑤ (9.0 h and 4.7 h knot gaps) and ⑦: cold NO₂ and reduced χ^2 per record, knot times as ticks. **(e)** Reported σ against residual RMS, hot ANs. **(f)** Fields flagging each episode.

8 Conclusions

We have presented Augur, an open DOAS retrieval posed in the extinction domain as a separable nonlinear least-squares problem and solved by variable projection. It is configured by instrument and campaign profiles, and its diagnostics were demonstrated on one field case: a thermal-dissociation broadband cavity-enhanced absorption spectrometer with two heated and one ambient-temperature cavity. The formulation is conventional; the implementation makes three things cheap: substituting a calibration reference without reprocessing the campaign, evaluating the projected objective anywhere in the nonlinear parameters, and carrying the termination state and the distance to each reference into the product.

The code solves the problem it states: synthetic recovery is at the level of the linear algebra, the analytic Jacobian agrees with finite differences to round-off, and Augur and QDOAS agree on synthetic spectra. On field spectra they agree in NO_2 with $r^2 \geq 0.990$ but differ in the heated-channel scale by 4–6 %. The full chain returns +0.0053 ppb at a true zero, a fortieth of the three-sigma detection limit of 0.220 to 0.257 ppb, which tests the additive chain only.

The first question concerned errors the fit cannot see. With white noise and a complete model the reported sigma tracks the synthetic scatter to within about ten per cent, rising to 1.32 with a free shift; against the calibration chain it fails. For the channel-difference product, perturbing the zero-air reference, the reflectivity and the etalon as the operational pipeline builds them gives, in the best-calibrated week, a structural contribution of 1.28 (robust scale) to 2.29 (standard deviation) times the reported sigma. The total uncertainty is 1.63 to 2.50 times what the fit reports for the same 60 s records. None of the calibration-chain terms enlarges the residual. These terms add to the campaign-constant cross-section and Rayleigh uncertainties of earlier treatments rather than replacing them (Sect. 4.6). The magnitudes belong to this instrument; the mechanism, an intermittently measured reference interpolated to every record, is general.

The second question concerned information the pipeline holds. With the cold shift bounded to ± 1 px in a sensitivity run, every base fit of the ambient-temperature channel returned its shift at the bound. In the budget propagation 187 of 9034 records were excluded because a fit terminated at a bound, and all were recorded as converged. In one window the projected objective varies by about one per cent across four pixels of shift, in another over five orders of magnitude, and both fits report convergence. A time-base error introduced by the pipeline, a reflectivity reference on a superseded time convention, displaced the alkyl nitrates by 0.100 ppb, while the residual and the reported uncertainty moved by two parts in a thousand.

A third category lies outside both. Against a co-located NIER analyser the heated-channel NO_2 reads low by a factor of about 0.6 while the ambient-temperature channel agrees. This multiplicative discrepancy, whose cause is unresolved, is the largest error in the record, and neither the fit report nor the proposed fields see it (Sects. 3.6 and 6.5).

We therefore propose that a retrieval report three numbers: the fit uncertainty, the structural term for the record's own calibration context, and a context flag (Sect. 6.1). Every implementation already computes the flag's fields, except the build time and time convention of a reference, which must be recorded when the reference is built. Quality gates should be reported with the coverage they cost, because they remove coverage as well as variance (Sect. 6.4). Where the interpolation error is set by knot noise, as in the 180 °C channel, averaging more scans per calibration block is the lever to test; the within-block evidence is unfavourable and the decisive measurement has not been made (Sect. 6.2).

The evidence is one instrument over one campaign. The code is not tied to that instrument (Appendix A), but its behaviour on other configurations is not shown here. The reporting convention and the synthetic benchmark of 261 cases, which contains no retrieval code or field data, should carry beyond it; extending the benchmark across line shapes, windows and pixel grids would separate what is general from what is ours.

730 Three extensions follow from the results. Fitting one shift jointly across the spectra of a calibration block could remove the indeterminacy diagnosed here, given an absorber in the block (Sect. 5.1). Treating the zero-air reference and the reflectivity as measured quantities with their own covariance would propagate the structural term inside the fit (Sect. 4.6). And the raw light level of each spectrum, the proximate cause of all six ambient-temperature excursions, should travel with the product together with the helium and zero-air intensities of each calibration block (Sect. 6.5).

735 *Code and data availability.* Augur is free software released under the MIT licence. The version used in this paper, v0.2.0, is archived at Zenodo ([https://doi.org/\[TBD-Z1\]](https://doi.org/[TBD-Z1])) and developed at <https://github.com/HwanKyungLee/CAESAR>. The archive contains the analysis code, the test suite referred to in Sect. 3, the scripts that produce the figures of this paper, and derived calibration constants (wavelength-calibration polynomials, instrument line-shape widths and instrument-convolved reference cross sections). The original laboratory cross sections are not redistributed; they are available from their providers: NO₂ (Vandaele et al., 2002), CHOCHO (Volkamer et al., 2005), O₄ (Thalman and
740 Volkamer, 2013), and H₂O line data from HITRAN (Gordon et al., 2022).

The synthetic benchmark of Sect. 3.4 – 261 spectra with known truth, their generator, the reference implementation and its scorecard – contains no field data and is archived separately ([https://doi.org/\[TBD-Z2\]](https://doi.org/[TBD-Z2])). Figure 3 can be reproduced from the two archives alone.

Raw instrument data and the field time series used in Sects. 4 to 7 (60 s extinction spectra, zero-air and reflectivity knot tables, and retrieved NO₂ amounts, 18 May to 11 July 2026) are not publicly distributed; they are available from the corresponding author on reasonable request.

745 **Appendix A: Instrument and retrieval configuration**

Augur reads the layout of the raw files – which columns hold which channel’s spectrum, which state flag marks an ambient, zero-air or helium scan, and where the housekeeping values sit – from a JSON instrument and campaign profile, and the fit configuration from a separate fit-setting file. The retrieval logic keys on the structure these files declare, not on channel names, so the same code is configured for another instrument or deployment by writing new files rather than by changing it. This
750 appendix collects the configuration of the field case of Sect. 7 in one place. Table A1 is transcribed from the operational fit-setting file of the campaign and the wavelength-calibration files it names; Table 1 is its short form.

References. The cross sections are NO₂ (Vandaele et al., 2002), CHOCHO (Volkamer et al., 2005) and H₂O from HITRAN line data (Gordon et al., 2022), each convolved with the line shape of its channel before the fit; O₄ is not fitted in the operational configuration. No robust weighting, regularisation or non-negativity constraint is applied (Appendix B6).

755 *Wavelength calibration.* Each channel has its own Hg-lamp calibration with a quadratic pixel-to-wavelength polynomial. Two calibrations of the 300 °C spectral region exist, one per region of interest on the detector; the operational fit setting uses

Table A1. Operational retrieval configuration of the three channels. “Linked” parameters take the value of the NO₂ parameter. The squeeze is stored as the departure from unity of s_i in Eq. (3), so 0 means no stretch. Dispersion is the median over the fitting window of the wavelength-calibration file; the line-shape width of the heated channels is the value recorded in that file (mean of four Hg lines), that of the ambient-temperature channel is the pair of measurements discussed in Sect. 4.1.

	300 °C	180 °C	ambient temperature
label in Sects. 3–6	hot ANs	hot PNs	cold
raw spectrum block	2053	4101	(cold file)
fitting window (nm)	429.5–461.9	444.1–470.6	438.4–475.8
baseline	Chebyshev, order 4	Chebyshev, order 3	Chebyshev, order 4
absorbers fitted	NO ₂ , CHOCHO, H ₂ O	NO ₂ , CHOCHO, H ₂ O	NO ₂ , CHOCHO, H ₂ O
NO ₂ shift (px)	bounded, −10 to +0.5	bounded, −5 to +5	fixed, −0.5
NO ₂ squeeze, $s - 1$	bounded, ±0.005	bounded, ±0.005	fixed, 0
CHOCHO, H ₂ O shift and squeeze	linked	linked	linked
cross-section temperature coefficient	0 (not applied)	0 (not applied)	0 (not applied)
κ_i			
negative amounts	allowed	allowed	allowed
optimiser step limit	on (0.5)	on (0.5)	on (0.5)
wavelength calibration	Hg, 19 June 2026, quadratic	Hg, 19 June 2026, quadratic	Hg (four lines), 23 May 2026, quadratic
dispersion (nm px ^{−1})	0.0484	0.0481	0.0482
line-shape FWHM (nm)	0.681	0.683	0.765–0.785

the second, the two differ by about 0.6 px (0.034 nm), and which of the two belongs to block 2053 is not settled. The difference is within the shift range of that channel and is absorbed by the fitted shift.

Temperature and pressure. Sample temperature and pressure are recorded with every spectrum and enter the Rayleigh correction of Eq. (1) and the conversion from number density to mixing ratio; their interpolation contributes below 0.1 % (Sect. 4.1). The temperature scaling $\tau_i(T)$ of Eq. (2) is not used ($\kappa_i = 0$).

Cavity. All three cavities have $d = 51.8$ cm and $R_L = 1.0$. R_L is assumed, not measured; since $\alpha \propto R_L$ it enters as a multiplicative factor (Sect. 3.6).

Calibration references. Zero-air and helium blocks are identical, 34 scans over 32 s; zero air is measured every hour and helium every three hours (Table A2). The zero-air reference I_0 is interpolated to the time of each spectrum from the neighbouring zero-air blocks, and the etalon frequency is held fixed (Sect. 4.1). Each reflectivity knot is formed from one zero-air block and the most recent helium block that passed the quality gates, so the knots of the heated channels are hourly – 1261 over the deployment at a median spacing of 1.00 h – while the helium half of a knot is up to three hours old; $R(\lambda, t)$ is the monotone piecewise-cubic interpolation of Sect. 2.4 through these knots, after the knot gate of Sect. 6.3.

Table A2. Calibration schedule of the heated channels, with block counts for the budget window (18–24 May): number of blocks, scans and duration per block, cadence and duty cycle, for zero air and helium.

	blocks	per block	cadence	duty
zero air	157	34 scans / 32 s	1.00 h	0.90 %
helium	54	34 scans / 32 s	3.00 h	0.31 %
				1.20 %

770 **Appendix B: The projected Jacobian, column scaling and the joint covariance**

B1 Setup

Write the forward model of Sect. 2.1 for one spectrum as

$$\alpha = \Phi(\theta)c + \varepsilon,$$

775 where $\theta = (\text{shift}, \text{squeeze})$ enters the columns of Φ through the resampling of the reference cross sections, and c collects the gas amounts and the Chebyshev baseline coefficients. The problem is separable: for any θ the linear parameters are determined by least squares, $c^*(\theta) = \Phi^+ \alpha$, and the objective reduces to the *profiled* residual

$$r(\theta) = \left(I - \Phi(\theta)\Phi(\theta)^+\right)\alpha = P^\perp(\theta)\alpha.$$

The optimiser searches only θ ; c is never a search direction. This is variable projection (Golub and Pereyra, 1973).

B2 The derivative has two terms

780 Differentiating the projector rather than the fit gives, for each nonlinear parameter θ_j and with $\Phi_j = \partial\Phi/\partial\theta_j$,

$$\frac{\partial r}{\partial\theta_j} = -\left[P^\perp\Phi_j\Phi^+ + \left(P^\perp\Phi_j\Phi^+\right)^\top\right]\alpha.$$

The first term is the change in the residual caused by moving the column space; the second is the change caused by the reprojection of α onto it. Kaufman (1975) drops the second, which is cheaper and is often adequate because it does not change the stationary point – both expressions vanish where the true gradient vanishes – only the path taken to it and the curvature reported there.

785 It is not adequate here. On a small separable problem with the same structure as Sect. 2.1 – two Gaussian absorber columns and three Chebyshev columns – the two-term expression agrees with a central finite difference of $r(\theta)$ to 2×10^{-10} relative, while the one-term expression is in error by 20 % and 17 % in the two nonlinear parameters. The step-size sweep of Sect. 3.2 measures the same thing on the real problem: the analytic derivative descends to the round-off floor, which an expression missing a term of that size could not do.

Table B1. Effect of rescaling one column of the small test problem of Appendix B2 on the condition number κ , on the projector $P^\perp$ and on the derivative $\partial r/\partial\theta$.

column scaling	$\kappa(\Phi D)$	$\ \Delta P^\perp\ $	relative change in $\partial r/\partial\theta$
none	9.3	0	0
10^{-3} on one column	6.1×10^5	2.4×10^{-10}	1.1×10^{-10}
10^{-8}	6.1×10^{11}	1.3×10^{-4}	5.1×10^{-5}
10^{-12}	7.7×10^{15}	1.00	0.48
10^{-19}	2.1×10^{16}	1.00	0.48

B3 Column scaling changes nothing, until it changes everything

Replace Φ by ΦD for any invertible diagonal D . Then $(\Phi D)^+ = D^{-1}\Phi^+$, so $P = \Phi D D^{-1}\Phi^+ = \Phi\Phi^+$ and $P^\perp$ are unchanged, and in the derivative of Appendix B2 the factors D and D^{-1} cancel term by term. The profiled residual, its derivative and the fitted θ are therefore invariant under rescaling of the columns; only c and its covariance rescale, and they rescale exactly
795 inversely, so the retrieved amounts are unchanged.

That is the algebra. The arithmetic is not invariant, and the failure is abrupt rather than gradual, because Φ^+ is computed with a rank tolerance (Table B1).

Once the scaling drives the condition number past the tolerance the pseudo-inverse discards the column, the projector is onto a different subspace, and the invariance is lost completely rather than degraded. The operational problem sits at the bottom of
800 that table by construction: a cross-section column is of order 10^{-19} cm^2 and a Chebyshev column is of order unity. Column normalisation is therefore not a numerical nicety but the condition under which the invariance of Appendix B3 holds at all, and the retrieval normalises every column to unit norm before the solve.

B4 The joint covariance and what the fit usually reports

Linearising the full problem in (c, θ) at the solution gives the Jacobian $J = [\Phi \ B]$ with $B = \partial(\Phi c)/\partial\theta$, and the covariance
805 $\sigma^2(J^\top J)^{-1}$. Partitioning it, the θ block is the Schur complement

$$\text{Cov}(\theta) = \sigma^2(B^\top P^\perp B)^{-1},$$

which is the projected curvature of Appendix B2 and not $\sigma^2(B^\top B)^{-1}$: the part of B that lies in the column space of Φ is absorbed by the linear parameters and constrains θ not at all. The c block is correspondingly larger than the $(\Phi^\top \Phi)^{-1}$ that a fit with θ held fixed would return, by an amount that depends on how much of each column's sensitivity the nonlinear parameters
810 can mimic. On the toy problem above the inflation runs from 1.001 to 1.257 across the five columns.

Both statements are elementary and neither is new. What matters for Sect. 5 is that a retrieval which reports $\sigma^2(\Phi^\top \Phi)^{-1}$ – the covariance of the linear parameters at a fixed θ – is reporting a quantity that is correct only if θ was known rather than fitted, and the difference is not small when θ is poorly determined. Augur forms the full $(J^\top J)^{-1}$ and reports both blocks;

the θ block is available at no extra cost once the projected Jacobian of Appendix B2 has been formed. It describes only the local curvature at the solution, which is why the identifiability statistics of Sect. 5 are taken from profile scans of the projected objective rather than from this block.

B5 Implementation of the projected Jacobian and relation to DOAS codes

In the implementation of Sect. 2.3 the weighted, column-scaled design matrix is factorised as $\mathbf{AS}^{-1} = \mathbf{QR}$, which delivers $\hat{c}$, the residual, and the projector $\mathbf{P}^\perp = \mathbf{I} - \mathbf{QQ}^\top$ from one factorisation, with $(\mathbf{A}^+)^\top = \mathbf{QR}^{-\top}$ in Eq. (6). Only the gas columns of $\mathbf{D}_k$ are non-zero: differentiating Eq. (3),

$$\frac{\partial}{\partial \delta_i} \sigma_i(p'_i) = \sigma'_i(p'_i), \quad \frac{\partial}{\partial s_i} \sigma_i(p'_i) = (p - p_c) \sigma'_i(p'_i), \quad (\text{B1})$$

so σ'_i is obtained once from the derivative of the interpolating spline that already represents each reference. The polynomial and etalon columns are independent of θ and contribute nothing. Both terms of Eq. (6) cost one triangular solve and one multiplication by $\mathbf{Q}$; their derivation and the invariance under the column scaling $\mathbf{S}$ are given in Appendices B2 and B3. The gradient of the objective is the same with and without the second term, because $\mathbf{A}^+ \mathbf{r} = 0$; the second term adds the exact curvature and makes the Jacobian testable against finite differences to round-off (Sect. 3.2). We have not compared the convergence rates of the two forms.

The separable structure is not new to DOAS; what the established treatments do not form is the derivative of the linear solution with respect to the nonlinear parameters. QDOAS solves the slant columns and the polynomial by singular value decomposition inside each evaluation of the fitting function while its Marquardt–Levenberg loop iterates over the nonlinear parameters only – the structure WinDOAS itself named “Marquardt-SVD” – but its source states that the derivatives with respect to shift and stretch are “always numeric”, so no term of the form of Eq. (6) is formed. DOASIS (Kraus, 2006) goes the other way: once shift and squeeze are fitted, the concentrations are handed to the nonlinear solver as well, and the thesis notes that the algorithm may then fail to find the global optimum that the linear step would have guaranteed. The standard textbook scheme (Platt and Stutz, 2008, Sect. 8.3.4) alternates one linear solve with one Levenberg–Marquardt step, explicitly “with fixed parameters a_j and c_h ” – block-coordinate optimisation, in which $\partial \hat{c} / \partial \theta$ does not appear. Outside DOAS, the analytic form of Eq. (6) is established in atmospheric remote sensing: Bärligea et al. (2023) use exactly this projected derivative, with the Kaufman variant, for a SWIR retrieval in which – as in BIRRA (Gimeno García et al., 2011) – the separability is inverted relative to Eq. (2).

For the covariance of Sect. 2.5: the inverse of the gas block of Eq. (8) is the Schur complement

$$\mathbf{A}_\mathbf{W}^\top \mathbf{A}_\mathbf{W} - \mathbf{A}_\mathbf{W}^\top \mathbf{V} (\mathbf{V}^\top \mathbf{V})^{-1} \mathbf{V}^\top \mathbf{A}_\mathbf{W},$$

which is never larger than the matrix inverted in Eq. (7), so the joint standard errors are never smaller than the conditional ones. $\mathbf{V}$ holds c fixed and is therefore not the projected Jacobian of Eq. (6); using Eq. (6) there would count the linear block twice.

B6 Scaling of the fitted quantity and options not used operationally

Equation (4) is solved on α scaled to order unity, and the linear coefficients are scaled back afterwards. Ambient α in these windows is of order 10^{-7} cm^{-1} , so residuals and gradients are of order 10^{-15} , while the default termination tolerances of a general-purpose least-squares routine are of order 10^{-8} . On unscaled α the termination test can then be satisfied at the starting point: the solver returns after a single function evaluation, having moved the shift by nothing, and reports success. In one operational spectrum the unscaled fit exited with shift $+0.0000 \text{ px}$ after $n_{\text{fev}} = 1$, while the same spectrum scaled by 10^7 converged to -0.5000 px after $n_{\text{fev}} = 5$. The failure is silent – it produces a plausible number with a success flag – and it is specific to re-implementations in scaled, tolerance-driven optimisers rather than to the DOAS method itself.

The linear step can be solved with a non-negativity constraint on the gas coefficients, and a Tikhonov penalty can be applied to the gas, etalon and auxiliary columns (never to the Chebyshev background, which would distort it). Neither is used in the configuration reported here: the operational λ is zero and the concentrations are left unconstrained so that values near zero remain unbiased. The analytic Jacobian of Eq. (6) requires the unconstrained linear step: at an active non-negativity bound the projector is not differentiable, and the implementation falls back to finite differences in that case. Robust (Tukey bisquare) iteratively reweighted fitting is available and likewise unused here.

Appendix C: Calibration-block length and duty cycle

This appendix gives the measurements behind the second paragraph of Sect. 6.2. They are not conclusive, for the reason stated below, and are recorded so that the measurement that would settle the question can be designed.

What acts on knot noise is the number of scans averaged into each calibration block. Splitting each block into contiguous quarters separates the two components that limit a block mean – white scan noise σ_s , which averages down, and drift within the block at rate g , which does not:

$$\begin{aligned} \text{scale}(Q_0 - Q_1)^2 &= \frac{8\sigma_s^2}{N} + \left(\frac{gT}{4}\right)^2, \\ \text{scale}\left(\frac{Q_0+Q_1}{2} - \frac{Q_2+Q_3}{2}\right)^2 &= \frac{4\sigma_s^2}{N} + \left(\frac{gT}{2}\right)^2. \end{aligned} \tag{C1}$$

Sub-blocks must be cut contiguously for this, because an interleaved (even/odd) split distributes neighbouring scans across both halves and cancels whatever they share. The interleaved split is kept as a control: the ratio between the two, at the same number of scans, measures how much of the block's scatter is common to neighbouring scans rather than independent.

The operational block is $N = 34$ scans spanning $T = 32.0 \text{ s}$, in both channels and for both reference gases. Within it, averaging does not behave as white noise (Table C1).

Doubling the number of scans averaged reduces the scatter by 7 % in the ANs channel, against the 29 % that independent scans would give. Most of what a block mean contains is therefore not independent between scans, and the last row says so directly: cutting the same block contiguously rather than in alternation more than doubles its apparent scatter in ANs. These are spectral-domain statistics: in the NO_2 amount, contiguous and interleaved halves give similar knot noise (Sect. 4.2), so most

Table C1. Scatter of contiguous sub-block contrasts within the operational calibration block ($N = 34, T = 32.0$ s) in the two heated channels, the averaging exponent they imply, and the ratio of contiguous to interleaved scatter at the same number of scans m , with the values expected for white noise.

	hot ANs	hot PNs	white noise
contiguous quarters (8 scans)	2.506×10^{-4}	1.095×10^{-4}	
contiguous halves (16 scans)	2.332×10^{-4}	0.820×10^{-4}	
averaging exponent	0.104	0.416	0.500
contiguous / interleaved, same m	2.26	1.22	1.03

Table C2. Calibration duty cycle for the operational zero-air block and for blocks two and four times longer.

zero-air block	N	zero-air duty	total calibration duty
operational	34	0.90 %	1.20 %
$\times 2$	68	1.79 %	2.10 %
$\times 4$	136	3.58 %	3.89 %

of this correlated structure is evidently absorbed by the baseline polynomial before it reaches the gas columns. A monotonic
875 trend across the 32 s is present in both channels but explains little of this – see below.

Two conclusions follow, and the second is a limitation. Within the range that can be measured, lengthening the averaging
window is worth much less than the $N^{-1/2}$ a reader would assume, and how much less depends on the channel. And the range
that can be measured is entirely below the operational block length: subdividing N scans says nothing about $2N$ or $4N$. The
evidence available is therefore not neutral but mildly unfavourable, and the measurement that would settle it – operating one
880 longer block – has not been made.

The duty cycle is not what stands in the way. Calibration occupies 1.2 % of the measurement time (Sect. 6.4), and a fourfold
zero-air block would take it to 3.9 % (Table C2).

Even under the white-noise assumption the return would be modest, and it is worth stating why before anyone spends duty
cycle on it. Halving the zero-air knot noise removes three quarters of the I_0 variance, but that term carries 45 % of the Σ ANs
885 variance, so the structural budget would fall from 0.1236 to 0.1008 ppb – 18.5 %, not half. The measured exponents put even
that out of reach in one channel; Table C3 gives the ceiling for each remedy.

These are ceilings, not forecasts, and the way they are computed needs stating because Sect. 4.3 forbids the obvious method.
Removing a term’s contribution from a total is a quadrature operation, and the joint propagation is not a quadrature sum –
here it exceeds one by 7.6 %, 0.1236 ppb against 0.1149. The rows are therefore evaluated in quadrature space and rescaled by
890 that measured ratio, which makes the operational row reproduce the measured joint exactly and applies the same correction to
the others. The residual assumption is that the ratio survives shrinking a term, which it will not do exactly, because the ratio
depends on the covariance and interaction of the terms (Sect. 4.3), and shrinking one term changes them.

Table C3. Best attainable Σ ANs structural uncertainty (standard deviation) under each remedy, as a fraction of the operational value, and the assumption each row rests on. These are ceilings, not forecasts.

	Σ ANs structural (SD), best attainable	of operational	status
operational	0.1236 ppb	1.00	measured
×4 blocks only	0.1008	0.82	assumes $N^{-1/2}$; not supported below N
flag bad R intervals only	0.0901	0.73	assumes a perfect detector
both	0.0547	0.44	both assumptions

The ratio column is unaffected by any of this: the rescaling factor cancels, so the relative ordering is the same whether the calculation is done in quadrature space or after rescaling. Section 6.2 uses only the ordering.

895 Flagging outranks averaging here, and it does so under every basis we have checked – record space as well as hourly, and quadrature space as well as rescaled. The flagging row assumes a detector that removes the tail and nothing else, reducing the R term to its present robust scale; a real flag will fall short. The block row assumes an averaging behaviour the measurements of this appendix do not support.

900 A linear trend across the block is part of the explanation but not all of it. It accounts for 40 % of the within-block variance in the ANs channel and 34 % in PNs, inflating the apparent sub-block scatter by 1.29 and 1.23. Whether removing it restores white averaging cannot be tested by detrending and re-measuring: the contrast comparing contiguous halves and the least-squares slope over four sub-blocks have a cosine of -0.894 , so subtracting the line subtracts most of the quantity being measured. Measured instead on a contrast orthogonal to the trend, the residual scatter is still 1.88 times white in ANs and 1.17 in PNs. Detrending is a partial remedy; what limits averaging is a correlation that outlives the removal of its linear part.

905 What does not depend on either assumption is the ordering. The reflectivity term is concentrated in a few episodes and does not average down; the zero-air term is distributed and averages at least partly. Flagging comes first whatever the detector achieves and whatever the block scaling turns out to be.

910 One observation from the same measurement is worth recording. Both channels show a monotonic trend across the 32 s block, of opposite sign – +248 to −314 ppm in ANs against −261 to +235 ppm in PNs. Two adjacent windows moving in opposite directions is a change of spectral slope rather than of intensity, which points at the source or at the bandpass separating the channels rather than at the gas fill. Removing the linear trend by contrast does not reduce the excess scatter, so this trend is not what limits averaging; the correlation between neighbouring scans is.

Table D1. Candidate denominators for the ratio of total to reported Σ ANs uncertainty: value (ppb), time resolution, and the ratio of the total uncertainty (Table 6, robust scale to standard deviation) to each. The zero-air value is from Sect. 7.2.

denominator	ppb	resolution	total / denominator
fit-propagated σ , per scan (used)	0.0541	60 s = 1 scan	1.63–2.50
uncertainty column shipped with the product	0.0806	5 min, systematic	1.32–1.83
fit-propagated σ , 5 min bin	0.025	5 min	not comparable
zero-air scatter, SD	0.0856	34-scan block means	not comparable

Appendix D: Supplementary budget analyses

D1 Choice of denominator

915 The ratio of total to reported uncertainty in Sect. 4.4 uses the per-scan fit-propagated σ . Three other quantities are available, and none of them answers the same question (Table D1).

The column shipped with the product is a gain systematic rather than a measurement uncertainty. Regressed against the channel NO₂ amount over 14 974 records it is 0.04209 [NO₂] + 0.00025 ppb with $r = +0.9997$, and the product header states that it covers day-to-day inter-channel gain only and must not be used alone to assess detection significance; it does not bound the absolute scale either (Sect. 3.6). Comparing the structural budget against it asks whether structure is comparable to the gain term, not whether it exceeds the precision the retrieval claims. The five-minute row is the fit uncertainty averaged over five scans – over the deployment it is 0.0558/ $\sqrt{5}$ – and dividing it into a structural scale measured at 60 s would mix resolutions in the opposite direction. The zero-air scatter is the precision floor, but it is measured on 34-scan block means, each retrieved against a reference interpolated across the removed block, so it combines a different scan count with reference-interpolation error and is not a single-scan quantity.

925 The per-scan fit uncertainty is the median, over 8832 of the budget records, of the NO₂ uncertainty written by the operational fit of each channel propagated through the channel difference, $\sqrt{\sigma_{300}^2 + g'^2 \sigma_{180}^2}$. Over the whole deployment, with the g' of each calibration state, the same median is 0.0558 ppb (total 1.59–2.43 times), and at five-minute resolution the fit uncertainty lies between 0.0217 and 0.0326 ppb (5th and 95th percentiles).

930 D2 Coupling of the structural terms to the wavelength shift

The structural terms of Sect. 4 differ in how strongly they couple to the shift (Table D2). An error in the reflectivity changes the magnitude of the extinction and little else: it moves the shift by a median of 0.006 px (90th percentile 0.027 px), although the largest displacement reaches 0.285 px. An error in the zero-air reference or in the etalon frequency changes the shape of the extinction and is partly absorbed by the wavelength axis. The shape statistic of Sect. 6.1.1 gives the same distinction independently: the reflectivity is a pure amplitude term, and the other two compete with the nonlinear parameters.

Table D2. Change in the fitted wavelength shift (px) when each structural term of Sect. 4 is perturbed alone, and when all three are perturbed together: median, 90th percentile and maximum over the 8847 clock-aligned budget records of Table 6 (for each Σ ANs record, the larger of the two channels’ changes).

perturbation	median	p_{90}	max
zero-air reference	0.187 px	0.826	4.388
reflectivity	0.006	0.027	0.285
etalon frequency	0.154	0.347	2.941
all three, joint	0.243	0.766	4.276

The same propagation shows that the budget of Sect. 4 is not an artefact of basin hopping in the shift. Records whose shift moved by more than a pixel under perturbation are 6.2 % of the total and carry 5.0 % of the variance, and removing them raises the budget (rSD 0.0693 to 0.0706 ppb, SD 0.1236 to 0.1244 ppb). These are the low-amplification records, where the absorption is deep and the retrieval well conditioned.

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